

Price Risk, Irreversible Investment, and the Welfare Benchmark for the Section 45Z Clean Fuel Production Credit

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We decompose the welfare case for the Section 45Z clean fuel production credit into its two statute-relevant components — the carbon externality and price risk under irreversible investment — and ask whether a flat per-gallon credit is the right instrument for the risk component. A two-stage stochastic mixed-integer program sites a forest-residue sustainable aviation fuel (SAF) supply chain across 1,374 southeastern U.S. counties under a kernel-density price distribution estimated on a constructed drop-in fuel price — slate-weighted wholesale gasoline–diesel plus the Renewable Fuel Standard credit value the fuel earns, weekly over 2013–2021; a closed-form benchmark calibrated to the same distribution supplies the analytics. Price risk justifies at most \$1.04/gal at any admissible risk weight — \$0.52/gal at the midpoint calibration — so risk alone cannot cover the statutory \$1.30–\$1.45/gal; the balance implies a social cost of carbon of \$77–\$91/tCO₂ under the as-enacted rate and flat-credit delivery. Committing capacity before prices are known costs \$0.30 B/yr in expected regret, concentrated in the lowest-price scenarios. On instrument form, a one-sided price floor restores the risk-neutral benchmark output at 69% of the flat credit’s expected outlay, and a two-sided contract-for-difference restores it at zero expected outlay.

Keywords: sustainable aviation fuel; stochastic programming; welfare analysis; CVaR; social cost of carbon; contracts-for-difference.

JEL codes: Q42, Q23, D81, H23, C61, R12.

1. Introduction

Sustainable aviation fuel (SAF) requires large, irreversible investments in feedstock logistics and refining capacity, yet future fuel prices remain highly uncertain. That combination makes the economics of SAF deployment fundamentally a problem of investment under uncertainty. In the southeastern United States, the region most often discussed as the likely platform for forest-residue-to-SAF deployment, the revenue price a producer would face — a drop-in fuel price inclusive of the Renewable Fuel Standard credits the fuel earns (Section 4.3) — has an estimated support of \$3.81–\$9.12/gal, while depot and biorefinery infrastructure must be committed before realized prices are observed.

U.S. policy now offers substantial per-gallon support for SAF through Section 45Z of the Inflation Reduction Act (IRA), with statutory credit values in the range of \$1.30–\$1.45/gal.¹ The statute scales the credit to carbon intensity, but the informal case for SAF support invariably adds a second argument: “de-risking” irreversible investment against volatile fuel prices. This paper prices both arguments in one framework. We decompose the capacity-restoring credit into a Pigouvian carbon component and a price-risk component, show the decomposition is exactly additive, and then ask a question the level-setting debate skips: whether a flat per-gallon credit is the right *instrument* for the risk component at all.

Existing work addresses parts of this problem but not the welfare question. Spatially detailed siting studies (Ekşioğlu et al. 2009; Lin et al. 2014; Marvin et al. 2012; Masum et al. 2023) represent feedstock heterogeneity, logistics, and processing costs richly, but typically treat uncertainty deterministically or through scenario analysis that does not generate true ex ante decisions. The stochastic-programming literature (Birge and Louveaux 2011; Schultz 2003; Shapiro, Dentcheva, and Ruszczyński 2014; Santos et al. 2019; Gebreslassie, Yao, and You 2012; Osmani and Zhang 2017; Quddus et al. 2018) provides the appropriate framework for here-and-now decision making under uncertainty, but is rarely implemented at regional spatial scale and rarely in a welfare setting. CVaR-based approaches (Rockafellar and Uryasev 2000, 2002; Artzner et al. 1999; Conejo, Carrión, and Morales 2010; Boomsma, Meade, and Fleten 2012; Insley 2017) offer a coherent framework for downside tail risk, yet they have not been integrated with a large-scale spatial siting problem under empirical price uncertainty. A separate strand compares renewable support instruments (feed-in tariffs, premia, and contracts-for-difference) under price uncertainty (Boomsma, Meade, and Fleten 2012), but that comparison has not been connected to the welfare accounting of a specific statutory credit. That connection is missing from the literature. The instrument-form result developed here supplies it.

We formulate SAF supply-chain investment as a social planner’s welfare-maximization problem with a make-or-buy choice between domestic SAF production and an outside option of wholesale fossil jet fuel at \$3.50/gal. We solve a two-stage stochastic mixed-integer linear program (SP-MILP) over 1,374 counties in the southeastern United States. Because a deep SAF spot market does not yet exist, the uncertain revenue price is a *constructed drop-in biofuel price* — a slate-weighted wholesale gasoline–diesel price plus the market value of the Renewable Fuel Standard credits the fuel generates, weekly from August 2013 through December 2021 — summarized by a kernel-density estimate (KDE). Its level embeds that credit value, so the price is

¹Section 45Z pays base rate $\times (50 - \text{CI})/50$ per gallon, with CI the pathway carbon intensity in kgCO₂e/mmBTU and a wage-and-apprenticeship-compliant SAF base rate of \$1.75/gal as enacted in the IRA (Congressional Research Service 2025b; Lohawala, Toman, and Joiner 2024). The \$1.30–\$1.45 range corresponds to a forestry-residue Fischer–Tropsch (FT) pathway CI of roughly 8.6–12.9 kgCO₂e/mmBTU under GREET-type accounting (Argonne’s Greenhouse gases, Regulated Emissions, and Energy use in Technologies life-cycle model).

policy-inclusive and the credit we price is incremental to the existing compliance stack (Section 4.3). We also derive a closed-form benchmark that maps the same price distribution into a welfare-equivalent per-gallon subsidy, and two propositions built on it: an exact additive decomposition of the capacity-restoring credit into Pigouvian and risk components (Proposition 1), and a strict expected-fiscal-outlay ranking of the instrument forms (flat credit, one-sided price floor, two-sided contract-for-difference (CfD)) that can deliver the risk component (Proposition 2).

Five findings emerge. First, for the committed central plan, the expected regret of acting before prices are known equals, for that fixed plan, the value of perfect information: \$299 M/yr, carried entirely by the nine lowest-price scenarios and vanishing above them. Second, that regret is tail-concentrated at the recorded solves (tail-to-mean 3.1, bounded within [1.1, 4.1] by residual solver gaps). The conditional value-at-risk $CVaR_{0.95}$ reaches \$921 M/yr, and evaluating the planner's mean-CVaR objective over the candidate plans splits the ranking at the risk weight: the committed chain wins expected welfare and both regret dimensions, but a plan one-fifth smaller carries a \$0.93 B/yr advantage in 5%-tail welfare, so a planner with risk weight above roughly 0.10 builds smaller. A fixed-demand chain-surplus analog of the same information question gives a candidate-based estimate of \$255 M/yr.

Third, calibrating the closed-form benchmark to the empirical price distribution yields a risk-premium subsidy of $\sigma^* = \$0.52/\text{gal}$, the per-gallon transfer that restores risk-neutral capacity for a mean-CVaR planner facing price risk and irreversible investment. That figure sits materially below the statutory \$1.30–\$1.45/gal Section 45Z credit, and the gap does not close at any admissible risk weight: the risk component tops out at \$1.04/gal even under maximal tail aversion. Fourth, adding the Pigouvian component (a life-cycle differential of 10.2 kgCO₂e/gal at the make-or-buy margin), the statutory credit is consistent with an implied social cost of carbon of \$77–\$91/tCO₂ under the baseline reading — as-enacted rate, flat-credit delivery, incremental to existing RFS support — between the two federal SCC vintages of \$51/tCO₂ (Interagency Working Group on Social Cost of Greenhouse Gases 2021) and \$190/tCO₂ (U.S. Environmental Protection Agency 2023). The implied SCC is a schedule in those choices rather than a point, and one row of the schedule is news: the post-2025 rate actually in force prices carbon at \$22–\$30/tCO₂, *below* the IWG vintage.

Fifth, the flat credit is the most expensive instrument for the risk component. A one-sided price floor at \$5.64/gal restores the same capacity at 69% of the flat credit's expected fiscal outlay (\$0.48 B/yr of expected outlay saved at the 3.0 B gal/yr anchor volume, worth up to \$0.14 B/yr in avoided deadweight loss when each dollar of public funds costs society \$1.30). A two-sided CfD at an actuarially fair strike restores it at zero expected outlay, moving the price risk from the committing region onto the federal balance sheet (Arrow and Lind 1970). The two results point the same way: pay less, and pay it differently.

The analysis is partial equilibrium. Section 6.1 signs the omitted channels one by one: financing frictions and investor hurdle wedges would raise the risk component, making the reported value a lower bound in those channels, while fixed-price offtake coverage, cost-price co-movement, and partial financial hedging would shrink it. The producer-side counterpart of the instrument-form result is developed in a companion paper (Li and Yu 2026). The remainder of the paper proceeds as follows. Section 2 develops the welfare framework and the closed-form benchmark, Section 3 presents the empirical stochastic-programming model, Section 4 describes the data and calibration, Section 5 reports the main results, Section 6 discusses sensitivity and limitations, and Section 7 concludes.

2. Conceptual Framework

A social planner chooses where to build a forest-residue-to-SAF supply chain in the southeastern United States, and at what scale, before SAF prices are realized. We organize the problem in four steps: the economic environment and timing (Section 2.1); the welfare objective with a make-or-buy outside option (Section 2.2); risk aversion and the value of information (Section 2.3); and a closed-form benchmark that maps the same primitives into a risk-premium-equivalent per-unit subsidy (Section 2.4). The closed form supports two further results used in Section 5: an exact additive decomposition of the capacity-restoring credit into a Pigouvian component and a risk component (Proposition 1), and an expected-fiscal-outlay ranking across the instrument forms that can deliver the risk correction (Proposition 2).

2.1. Economic environment and timing

Let $\mathcal{H}$ index residue-source counties, $\mathcal{D}$ candidate *depot* counties (with biomass pre-processing capability), and $\mathcal{R}$ candidate *biorefinery* counties. A scenario $\omega \in \mathcal{S}$ realizes a SAF wholesale price $p_{\text{SAF}}(\omega)$ and a feedstock-quality shock $\xi(\omega)$, with KDE-implied probability p^ω (proxied by historical wholesale jet-fuel spot prices, U.S. Energy Information Administration 2024). Following Li et al. (2026), we discretize feedstock ash content into percentile-based tiers $\mathcal{A}_d, \mathcal{A}_r$ at depots and refineries, with the SAF yield at refinery ash configuration $a' \in \mathcal{A}_r$ written $\phi(a'; \omega)$.

Decisions arrive in two stages. In Stage 1, the planner commits binary facility variables $y \in \{0, 1\}^{|\mathcal{D}|+|\mathcal{R}|}$. The scenario ω then realizes, and in Stage 2 the planner chooses continuous residue and pre-processed flows $\mathbf{x}(\omega), \mathbf{z}(\omega) \geq 0$ together with imported jet-fuel slack $s(\omega) \geq 0$. The Stage 1 choice is therefore made here-and-now under uncertainty; Stage 2 captures flow-level recourse after ω is observed.

2.2. Welfare objective and make-or-buy structure

Let $Q \in \mathbb{R}_{\geq 0}$ denote the year-zero representative output (gallons of SAF). For an inverse demand $P^d(Q; \omega)$, the *consumer surplus* is

$$(1) \quad \text{CS}(Q; \omega) = \int_0^Q P^d(q; \omega) dq - P^d(Q; \omega) \cdot Q.$$

Producer surplus is revenue minus all chain costs:

$$(2) \quad \text{PS}(\mathbf{x}, \mathbf{z}, y; \omega) = p_{\text{SAF}}(\omega) \cdot Q(\mathbf{z}; \omega) - C_{\text{tot}}(\mathbf{x}, \mathbf{z}, y),$$

where $Q(\mathbf{z}; \omega) = \sum_{(d,a,r,a')} \beta \cdot \phi(a'; \omega) \cdot z_{d,a,r,a'}$ is total SAF output and C_{tot} stacks stumpage, harvest, transport, and facility opex and capex. In the endogenous-demand variant the realized price is the demand height at the equilibrium quantity, $p_{\text{SAF}}(\omega) = P^d(Q(\omega); \omega)$, and $\text{CS} + \text{PS}$ is total surplus; in the fixed-demand model P^d is horizontal, $p_{\text{SAF}}(\omega)$ is exogenous, and the identification is replaced by the demand floor with slack imports (Section 3.5 states which results use which).

Make-or-buy structure. When domestic chain output falls short of demand, the gap is met by imported wholesale jet fuel at price p^{jet} (a perfect substitute within ASTM blend limits). Letting $s(\omega) \geq 0$ denote slack

imports, the social opportunity cost of slack is $p^{\text{jet}} \cdot s(\omega)$; we calibrate $p^{\text{jet}} = \$3.50/\text{gal}$ as a long-run benchmark from 2014–2024 U.S. Energy Information Administration (EIA) Gulf-Coast jet-fuel data, with sensitivity to the lower 2025 range reported in Section 6. The outside option is essential to the welfare framing: it assigns an economically meaningful opportunity cost to unmet SAF demand and supports the build-vs-no-build decision.

The planner’s problem is

$$(3) \quad \max_{\substack{y \in \{0,1\}^{|\mathcal{D}|+|\mathcal{R}|} \\ \mathbf{x}, \mathbf{z}, s \geq 0}} \mathbb{E}_{\omega} \left[\text{CS}(\mathbf{z}; \omega) + \text{PS}(\mathbf{x}, \mathbf{z}, y; \omega) - p^{\text{jet}} \cdot s(\omega) \right],$$

subject to mass-balance, capacity, quality, and demand-satisfaction constraints stated in Section 3.

remark (Carbon externality). *On a consistent life-cycle basis, the fossil-jet baseline of CORSIA (the Carbon Offsetting and Reduction Scheme for International Aviation), 89 gCO₂e/MJ (megajoule) (International Civil Aviation Organization 2019), converts at 131.5 MJ/gal (lower heating value) to 11.7 kgCO₂e/gal, while residue-derived Fischer–Tropsch SAF emits 1.1–1.5 kgCO₂e/gal (8.3 gCO₂e/MJ at the CORSIA default core life-cycle-assessment (LCA) value for forestry-residue FT; up to ~11 gCO₂e/MJ in GREET-type accountings, Argonne National Laboratory 2024). The life-cycle differential is therefore $\Delta e \approx 10.2\text{--}10.6 \text{ kgCO}_2\text{e/gal}$; we use the conservative $\Delta e = 10.2$ throughout. The Pigouvian internalization term $\text{ENV}(\mathbf{x}, \mathbf{z}, y; \omega) = -p_C \cdot E(\mathbf{x}, \mathbf{z}, y; \omega)$ (where $E(\cdot)$ denotes the chain’s life-cycle CO₂e emissions and p_C the carbon price) and the IRA Section 45Z producer credit are both deferred; we set $p_C = 0$ in the stochastic program so that the price-uncertainty channel is isolated, and price the carbon channel analytically in Section 5.5.*

remark (Welfare framing versus producer NPV). *At the price-taker limit $P^d(Q; \omega) = p_{\text{SAF}}(\omega)$ and $p_C = 0$, problem (3) reduces to a producer-side net-present-value (NPV) problem of the kind studied by Ekşioğlu et al. (2009); Lin et al. (2014); Marvin et al. (2012), except for the make-or-buy term $-p^{\text{jet}} \cdot s(\omega)$, a welfare quantity that makes the build-vs-no-build decision non-trivial. The welfare framing also delivers the closed-form public-finance object σ^* derived in Section 2.4, which the producer-NPV reduction does not support.*

2.3. Risk aversion and information value

Mean–CVaR welfare. If the planner evaluating the (regional) welfare aggregate is risk-averse over the tail of its distribution at confidence level α (the agent-identity conventions are fixed in Section 2.4), the expectation in (3) is replaced by the mean–CVaR functional

$$(4) \quad (1 - \lambda) \cdot \mathbb{E}[W] + \lambda \cdot \text{CVaR}_{\alpha}[W], \quad \lambda \in [0, 1],$$

where $W = \text{CS} + \text{PS} + \text{ENV}$ is total welfare. The functional is coherent in the sense of Artzner et al. (1999), and admits the Rockafellar and Uryasev (2000) LP linearization, which we state for completeness; the empirical solves of Section 3 are risk-neutral, and the mean–CVaR program is not solved on this instance (Section 5.8). The CVaR functional in (4) and the empirical regret statistic reported below price different things and should not be conflated. The functional prices tail risk in *welfare levels* (loss $-W$, whose bad tail is the low-price state), while the empirical results of Section 5 instead report per-scenario *regret* relative to the wait-and-see ideal, $L_s = W_s^{\text{WS}} - W_s^{\text{HN}}$, which turns out to concentrate in the same low-price states (over-committed capacity). Section 5.2 explains how the two relate.

Value of information. The information question rests on three benchmark welfare values. Let $W^*(\omega) = \max W(\mathbf{x}, \mathbf{z}, y; \omega)$ be the per-scenario wait-and-see optimum, W^{RP} the two-stage recourse-problem optimum with y shared across ω , and W^{EEV} the expected-value-of-expected-value objective evaluated when y is fixed at its mean-scenario optimum and $(\mathbf{x}, \mathbf{z})$ adapt. The *expected value of perfect information* and the *value of the stochastic solution* are

$$(5) \quad \text{EVPI} = \mathbb{E}[W^*(\omega)] - W^{\text{RP}}, \quad \text{VSS} = W^{\text{RP}} - W^{\text{EEV}}.$$

EVPI is the planner's willingness to pay for perfect SAF-price forecasting; VSS is the cost of replacing the scenario distribution with its mean. Both are reported in Section 5.

2.4. Closed-form benchmark: risk-premium-equivalent subsidy, decomposition, and instrument form

The large-scale program in (3) delivers a numerical optimum but leaves the economic mechanism implicit. We close this gap with a two-state reduction that admits a closed-form solution, isolating how the welfare-equivalent per-unit SAF subsidy depends on the upside probability of high demand, the demand-state spread, and the planner's risk-aversion weight. Two further results follow from the same machinery: an exact additive decomposition of the capacity-restoring credit into a Pigouvian component and a risk-premium component (Proposition 1), and a strict expected-fiscal-outlay ranking across the instrument forms that can deliver the risk correction (Proposition 2).

Setup. Reduce (3) to a single output decision $Q \in \mathbb{R}_+$ in a two-state economy with demand state $\omega \in \{H, L\}$ and probabilities (π_H, π_L) summing to one. State-conditional inverse demand is linear, $P^d(Q; \omega) = a(\omega) - bQ$ with $b > 0$, and the intercept $a(\omega) \in \{a_H, a_L\}$ captures high/low realizations of the SAF willingness-to-pay distribution. Marginal supply cost is a constant c , and the representative-investor capacity cost is $C(Q) = \frac{k}{2}Q^2$ with $k > 0$. Writing $K \equiv b + k$ for the combined curvature, the area-under-demand-minus-cost integral gives

$$(6) \quad W(Q; \omega) = (a(\omega) - c)Q - \frac{K}{2}Q^2,$$

and the mean-CVaR functional in (4) applies directly.

Assumptions. The following maintained assumptions are used throughout this subsection and the proofs in Section A.

assumption 1 (Two-state demand). $\omega \in \{H, L\}$ with $\pi_H, \pi_L > 0$ and $\pi_H + \pi_L = 1$.

assumption 2 (Linear inverse demand). $P^d(Q; \omega) = a(\omega) - bQ$ with $b > 0$.

assumption 3 (Convex capacity cost). $C(Q) = \frac{k}{2}Q^2$ with $k > 0$.

assumption 4 (State ordering). $a_H > a_L$.

assumption 5 (Positive net premium). $a_L > c$: even in the bad state the planner finds it worthwhile to produce.

assumption 6 (CVaR confidence). $\alpha \in (0, 1)$. The closed-form $\sigma^*(\alpha)$ is piecewise on $(0, \pi_H)$ and $[\pi_H, 1)$.

assumption 7 (Bounded risk-aversion weight). $\lambda \in [0, 1]$.

assumption 8 (Partial equilibrium). $a(\omega)$ is treated as an exogenous random variable; no general-equilibrium substitution between SAF and fossil jet fuel is modeled.

assumption 9 (External financing). Policy transfers are financed from outside the welfare aggregate over which the mean-CVaR functional (4) is evaluated: a credit enters the marginal return of the committing agent but its funding does not enter the CVaR argument.

Assumption 9 fixes whose risk the functional prices and who pays for the correction, and it makes a transfer capable of moving the optimum at all: were the credit financed *inside* the aggregate, it would net out state by state and leave the program unchanged. Two readings satisfy the assumption. In the regional reading, the mean-CVaR functional is evaluated over the Southeast regional aggregate and the credit is a federal transfer into a small open region, the reading that matches the spatial scope of the empirical model in Section 3. In the decentralized reading, the tail-averse agent is the investor committing irreversible capacity, evaluated against the risk-neutral planner’s target output, the reading developed in the companion paper (Li and Yu 2026). The algebra below is identical under either reading; we use “the committing agent” where the distinction matters.

The assumption also fixes the normative benchmark. The welfare criterion against which σ^* is a correction is the risk-neutral objective $\mathbb{E}[W]$ of the external financier (the federal side, whose taxpayer pooling supports risk neutrality); λ describes the risk-bearing capacity of the committing agent, not the social planner’s preference. σ^* therefore prices a *jurisdictional risk-bearing wedge* — the cost of concentrating an irreversible, price-exposed investment on a regional or project balance sheet — rather than a Pigouvian market failure. Were λ instead a genuine social preference, Q_{risk}^* would itself be the optimum and no correction would be warranted.

Microfoundation. Assumption 9 is the reduced form of an incomplete-markets friction, not a preference assumption. The state-contingent claims that would let the committing agent sell its low-price exposure do not exist at the relevant tenor: liquid hedging tenor in the heating-oil/jet complex runs two to three years against a ten-year operating commitment on a twenty-year asset; the premium of the produced fuel over jet-A has no settlement index and is therefore pure basis risk; and a single-asset project company faces margin-call liquidity constraints that make a decade-long short position infeasible at scale (Froot, Scharfstein, and Stein 1993; Holmström and Tirole 1998). In the complete-markets benchmark λ would be irrelevant and $\sigma^* = 0$; $\lambda > 0$ measures the shadow price of the missing claims on the committing balance sheet, so the committing agent’s allocation is constrained inefficient relative to the risk-neutral financier’s feasible set, and a correction can raise welfare in that constrained sense — this is the sense in which we call the corrective credit “capacity-restoring” below. The Arrow and Lind (1970) pooling argument covers the financier’s side of the transfer; the friction above is what keeps the committing side from replicating it privately. The corresponding first-best is to *complete the market* — Proposition 2(iii)’s two-sided contract — rather than to transfer resources; the flat credit’s σ^* serves as the level-equivalent benchmark against which the statute is read, not as the recommended instrument.

Main result.

LEMMA 1 (Strict concavity and interior optimum). Under Assumptions 1–7, the mean-CVaR welfare functional (4) evaluated at the per-state welfare (6) is strictly concave in Q on $\mathbb{R}_+$ with second derivative $-K < 0$, and the maximizer Q_{risk}^* is interior. Proof in Section A.

THEOREM 1 (Risk-premium-equivalent subsidy). *Under Assumptions 1–9, the per-unit social subsidy σ^* that restores the risk-neutral output level Q_{RN}^* is*

$$(7) \quad \sigma^*(\alpha) = \lambda \cdot (a_H - a_L) \cdot \min\left\{\frac{\alpha \pi_L}{1 - \alpha}, \pi_H\right\}.$$

In the collapsed regime $\alpha \geq \pi_H$, which includes the baseline $\alpha = 0.95$ at $\pi_L \approx 0.5$, the expression simplifies to

$$(8) \quad \sigma^* = \lambda \pi_H (a_H - a_L).$$

Proof in Section A.

COROLLARY 2 (Expected value of perfect information). *Under Assumptions 1–8, the EVPI for the risk-neutral planner equals*

$$(9) \quad \text{EVPI} = \frac{\text{Var}(A)}{2K} = \frac{\pi_H \pi_L (a_H - a_L)^2}{2K}.$$

Proof in Section A.

COROLLARY 3 (Continuity at the regime boundary). *$\sigma^*(\alpha)$ is continuous on $(0, 1)$, with a kink in the derivative but no jump in the value, at $\alpha = \pi_H$. Proof in Section A.*

Economic interpretation. The collapsed-regime expression (8) admits a clean three-factor reading: $\sigma^* = \lambda \cdot \pi_H \cdot (a_H - a_L)$ is the product of the planner’s risk-aversion weight, the upside probability, and the demand-state spread. Each factor maps to an observable: λ is the committing agent’s risk-bearing primitive — the weight a regulator selects when evaluating the exposed jurisdiction’s tail (mean–CVaR welfare reduces to expected welfare at $\lambda = 0$ and to worst-tail welfare at $\lambda = 1$); π_H is the upside probability calibrated from the SAF-price KDE; and $a_H - a_L$ is the conditional-mean spread of the two-state median split of the KDE.

The result extends the prices-versus-quantities benchmark of [Weitzman \(1974\)](#) to a mean–CVaR social planner, and is the discrete-scenario mean–CVaR counterpart of the closed-form feed-in tariff derived by [Boomsma, Meade, and Fleten \(2012\)](#) in a continuous-time risk-neutral real-options model. At $\lambda = 0$ the closed form recovers Weitzman’s no-subsidy benchmark $\sigma^* = 0$.

Extension 1: additive Pigouvian decomposition. The statutory Section 45Z credit is designed as a carbon-intensity-scaled production credit, so any welfare accounting of its level must handle the carbon externality and the risk premium jointly. In the mean–CVaR framework the two components are exactly additive.

PROPOSITION 1 (Additive credit decomposition). *Suppose each unit of output carries an external social benefit $v \geq 0$ (the monetized value of avoided life-cycle emissions) not reflected in the market intercept $a(\omega)$. Under Assumptions 1–9: (i) the first-best output is $Q_{\text{FB}}^* = (\mathbb{E}[A] + v)/K$; (ii) the per-unit credit that moves the mean–CVaR planner to Q_{FB}^* is*

$$(10) \quad \sigma_{\text{tot}}^* = v + \sigma^*(\alpha),$$

with $\sigma^(\alpha)$ exactly as in Theorem 1; and (iii) the decomposition has no interaction term: the risk component is*

invariant to v , and the Pigouvian component is invariant to $(\lambda, \alpha, \pi_H, a_H - a_L)$. Proof in Section A.

Additivity is a consequence of the linearity of the effective intercept A_{eff} in the state-independent shift v : a constant per-unit externality raises the expectation and the tail conditional mean by the same amount, so it moves the target output without changing the risk wedge. Proposition 1 licenses the empirical decomposition of Section 5.5: the Pigouvian component can be priced off the life-cycle emissions differential and a social cost of carbon without re-solving the stochastic program, and added to the KDE-calibrated σ^* without double counting.

Extension 2: instrument form. Theorem 1 prices the risk correction as a *flat* per-unit credit — the instrument form of the statutory 45Z. But a flat credit shifts the mean of the payoff distribution while leaving its variance intact, whereas the distortion it corrects is a variance phenomenon. Instruments that operate directly on the variance deliver the same output correction at strictly lower expected fiscal outlay.

PROPOSITION 2 (Instrument form and expected fiscal outlay). *Let $m_\alpha \equiv \min\{\alpha\pi_L/(1-\alpha), \pi_H\}$, so that $\sigma^*(\alpha) = \lambda(a_H - a_L)m_\alpha$ by Theorem 1. Under Assumptions 1–9, each of the following instruments restores the risk-neutral output $Q_{\text{RN}}^* = \mathbb{E}[A]/K$, with expected per-unit fiscal outlay T as stated:*

(i) Flat credit at rate $\sigma^*(\alpha)$, paid in every state: $T_{\text{flat}} = \sigma^*(\alpha)$.

(ii) One-sided price floor (CfD-with-floor) at strike

$$(11) \quad \kappa^* = \frac{\pi_L a_L + \lambda m_\alpha a_H}{\pi_L + \lambda m_\alpha} \in (a_L, a_H),$$

paying $\kappa^* - a(\omega)$ per unit only when $a(\omega) < \kappa^*$:

$$(12) \quad T_{\text{floor}} = \sigma^*(\alpha) \cdot \frac{\pi_L}{\pi_L + \lambda m_\alpha} < T_{\text{flat}}.$$

(iii) Two-sided contract-for-difference at strike $\bar{a} \equiv \mathbb{E}[a(\omega)]$, exchanging the stochastic margin for the deterministic one ($\bar{a} - a(\omega)$ per unit, negative in the high state): $T_{\text{CfD}} = 0$, for every $\lambda \in [0, 1]$ and $\alpha \in (0, 1)$.

For $\lambda \in (0, 1]$ the ranking is strict: $0 = T_{\text{CfD}} < T_{\text{floor}} < T_{\text{flat}} = \sigma^*(\alpha)$. Proof in Section A.

COROLLARY 4 (Welfare wedge under costly public funds). *At marginal cost of public funds $\mu = 1$ the three instruments of Proposition 2 are welfare-equivalent (transfers net out of the planner's objective), and the ranking concerns expected public outlay only. At $\mu > 1$ the outlay differential becomes a strict welfare wedge: instrument i dominates instrument j by $(\mu - 1)(T_j - T_i) Q_{\text{RN}}^*$ per period. Proof in Section A.*

The economics of Proposition 2 is Arrow–Lind risk reallocation (Arrow and Lind 1970): the two-sided CfD transfers the price risk from the tail-averse committing agent (Assumption 9) to the external financier, so the risk wedge disappears without any expected transfer; the one-sided floor concentrates payments in exactly the low states that drive the CVaR term, which is why it undercuts the flat credit by the factor $\pi_L/(\pi_L + \lambda m_\alpha)$; and the flat credit, which pays identically in good and bad states, must purchase the entire risk premium at par. The conditions under which the external financier can absorb the swapped risk at zero premium, and the qualifications when the risk is macro-correlated, are discussed with the empirical calibration in Section 5.6. The producer-side counterpart of this ranking, where the CfD-with-floor relaxes a project-finance participation constraint at minimum expected transfer, is developed in the companion paper (Li and Yu 2026); the result here

Table 1: Comparative statics for σ^* and EVPI in the collapsed regime $\alpha \geq \pi_H$.

Partial	Closed form	Sign	Interpretation
$\partial\sigma^*/\partial\lambda$	$\pi_H (a_H - a_L)$	+	Higher committing-agent risk aversion lifts the subsidy.
$\partial\sigma^*/\partial\pi_H$	$\lambda (a_H - a_L)$	+	Higher upside probability lifts the subsidy.
$\partial\sigma^*/\partial(a_H - a_L)$	$\lambda \pi_H$	+	Wider demand spread lifts the subsidy.
$\partial\sigma^*/\partial\alpha$ (split)	$\lambda \pi_L (a_H - a_L)/(1 - \alpha)^2$	+	Below the kink, tighter tail focus lifts the subsidy.
$\partial\sigma^*/\partial\alpha$ (coll.)	0	0	Above the kink, the subsidy is α -independent.
$\partial\text{EVPI}/\partial \text{Var}(A)$	$1/(2K)$	+	More demand uncertainty raises information value.
$\partial\text{EVPI}/\partial K$	$-\text{Var}(A)/(2K^2)$	-	Steeper curvature dampens information value.

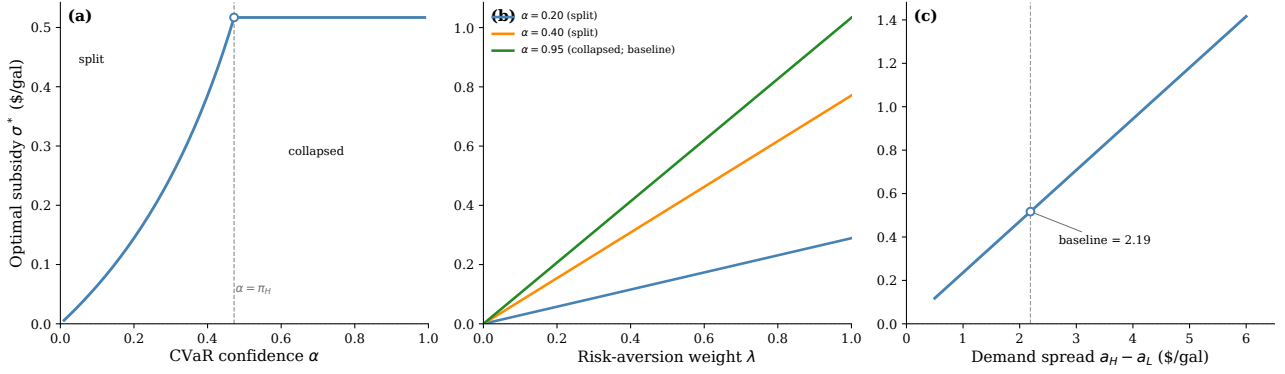


Figure 1: Comparative statics of σ^* . (a) $\sigma^*(\alpha)$, continuous at the regime boundary $\alpha = \pi_H$ with a kink in the derivative (Corollary 3). (b) $\sigma^*(\lambda)$ at three CVaR confidence levels. (c) $\sigma^*(a_H - a_L)$ at the baseline $(\lambda, \alpha, \pi_H) = (0.5, 0.95, 0.472)$, with the KDE-anchored spread \$2.19/gal marked.

needs only Assumption 9 and the mean-CVaR objective already in place.

Bridge to the empirical calibration. The two-state primitives map to the empirical KDE on SAF prices (Table 2, Section 4) via a two-state collapse at the bin-8/9 boundary (\$5.93, the bin edge nearest the binned KDE mean $\overline{p}_{\text{SAF}} = \$6.00/\text{gal}$; the raw sample mean is \$5.97, used with the deterministic siting solve of Section 5.1): $\pi_H = 0.472$ above and $\pi_L = 0.528$ below, with conditional means $a_H = \$7.16/\text{gal}$ and $a_L = \$4.97/\text{gal}$. One assumption in this mapping deserves emphasis: the conditional means are moments of the *realized price*, while the closed form’s $a(\omega)$ is a demand intercept. The bridge is the parallel-shift scenario mapping of Section 3.5, under which a price realization shifts the willingness-to-pay intercept one-for-one, so realized price dispersion is treated as dispersion in the state-conditional net premium $A(\omega) = a(\omega) - c$. The closed form is agnostic between demand- and cost-side sources of that dispersion *provided the two do not covary*; positive cost-price covariance would compress the net-premium spread and lower σ^* , the natural-hedge channel discussed in Section 6.1. At the baseline $\lambda = 0.5$, (8) delivers $\sigma_{\text{KDE}}^* \approx \$0.52/\text{gal}$, the per-unit policy number anchored on the empirical price distribution. Table 1 summarizes the comparative statics, and Figures 1 and 2 plot σ^* and EVPI against their primitives at the baseline calibration; numerical results for the large-scale SP-MILP are reported in Section 5.

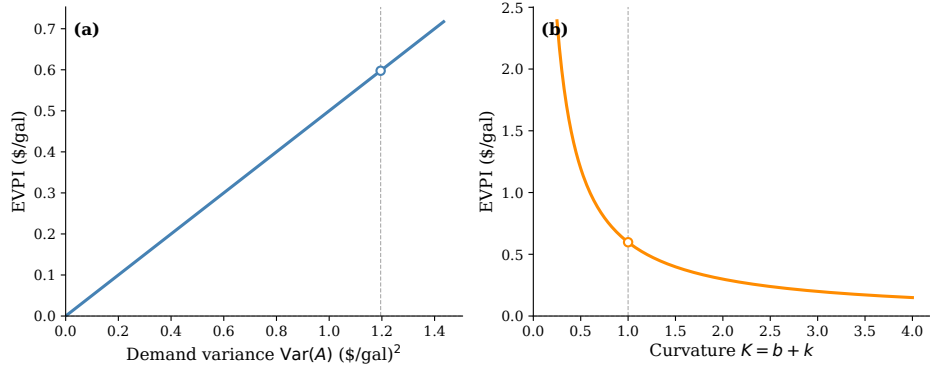


Figure 2: Sensitivity of EVPI to its two primitives. (a) EVPI against $\text{Var}(A)$ via the demand-state spread. (b) EVPI against the combined curvature $K = b + k$. Both panels plot the closed form (9) directly at the baseline calibration.

3. Empirical Model

We discretize (3) into a two-stage stochastic mixed-integer linear program, with first-stage facility binaries shared across scenarios.

3.1. Decision variables

- $\text{DvD}_{d,a} \in \{0, 1\}$: depot at county d in ash category a , $d \in \mathcal{D}$, $a \in \mathcal{A}_d$.
- $\text{DvB}_{r,a'} \in \{0, 1\}$: refinery at county r in ash category a' , $r \in \mathcal{R}$, $a' \in \mathcal{A}_r$.
- $x_{h,d,a}^\omega \geq 0$: residue (dt, dry tons) from h to depot (d, a) in scenario ω .
- $z_{d,a,r,a'}^\omega \geq 0$: pre-processed flow depot $(d, a) \rightarrow$ refinery (r, a') in scenario ω .
- $\text{slack}^\omega \geq 0$: imported SAF in scenario ω .

3.2. Constraints

Supply. For all $h \in \mathcal{H}$, $\omega \in \mathcal{S}$:

$$(13) \quad \sum_{d \in \mathcal{D}} \sum_{a \in \mathcal{A}_d} x_{h,d,a}^\omega \leq \bar{R}_h,$$

where $\bar{R}_h$ is the Stage-1-derived available residue (county aggregation: $\bar{R}_F = \sum_h \theta_{h,F} \cdot \bar{R}_h^{\text{hex}}$, with $\theta_{h,F}$ the hex-to-county area share; see Section B).

Depot capacity & ash bounds. For all (d, a) , ω :

$$(14) \quad \sum_h x_{h,d,a}^\omega \cdot \text{Res2Pre} \geq \text{DvD}_{d,a} \cdot \kappa_{\text{lw}}^D,$$

$$(15) \quad \sum_h x_{h,d,a}^\omega \cdot \text{Res2Pre} \leq \text{DvD}_{d,a} \cdot \kappa_{\text{up}}^D,$$

$$(16) \quad \sum_h \text{AshStd}_h \cdot \xi(\omega) \cdot x_{h,d,a}^\omega \geq \text{DvD}_{d,a} \cdot \kappa_{\text{lw}}^D \cdot \pi_a^D,$$

$$(17) \quad \sum_h \text{AshStd}_h \cdot \xi(\omega) \cdot x_{h,d,a}^\omega \leq \text{DvD}_{d,a} \cdot \kappa_{\text{up}}^D \cdot \bar{\pi}_a^D,$$

$$(18) \quad \sum_a \text{DvD}_{d,a} \leq 1.$$

Depot mass balance & refinery capacity / ash. For all (d, a) and (r, a') :

$$(19) \quad \sum_{r,a'} z_{d,a,r,a'}^\omega = \text{Res2Pre} \sum_h x_{h,d,a}^\omega,$$

$$(20) \quad \sum_{d,a} \beta \cdot \phi(a') \cdot z_{d,a,r,a'}^\omega \geq \text{DvB}_{r,a'} \cdot \kappa_{lw}^R,$$

$$(21) \quad \sum_{d,a} \beta \cdot \phi(a') \cdot z_{d,a,r,a'}^\omega \leq \text{DvB}_{r,a'} \cdot \kappa_{up}^R,$$

$$(22) \quad \sum_{a'} \text{DvB}_{r,a'} \leq 1,$$

$$(23) \quad \sum_{d,a} \text{DAshMean}_a \cdot z_{d,a,r,a'}^\omega \geq \frac{\kappa_{up}^R}{\beta} \cdot \text{DvB}_{r,a'} \cdot \{\pi_{a'}^R, \bar{\pi}_{a'}^R\}.$$

Here $\text{Res2Pre} = 0.85$ is the depot pretreatment recovery (the fraction of delivered residue that survives pre-processing), AshStd_h is the standardized ash content of county h 's residue, DAshMean_a is the mean ash content of depot ash class a , and the κ and π ($\bar{\pi}$) constants are capacity and ash-share lower (upper) bounds at depots (D) and refineries (R).

Market clearing. For all ω :

$$(24) \quad \text{slack}^\omega + \sum_{d,a,r,a'} \beta \cdot \phi(a') \cdot z_{d,a,r,a'}^\omega \geq \bar{Q},$$

where β denotes the biofuel conversion yield (gal of SAF per dry ton of residue).

3.3. Objective: Discretized social welfare

The discretized form of (3) (assuming horizontal demand, risk-neutral; the endogenous-demand variant used for the regret analysis is specified in Section 3.5, and risk aversion in Section 2.3) is as follows. Under horizontal demand the consumer-surplus integral in (1) is identically zero, so this fixed-demand objective is the chain surplus (producer surplus net of the slack opportunity cost), which is the accounting behind Result 2.

$$(25) \quad \max_{\omega} \frac{1}{1+\rho} \sum_{\omega} p^\omega \left[p_{\text{SAF}}(\omega) \sum_{d,a,r,a'} \beta \phi(a') z_{d,a,r,a'}^\omega - \text{Stump}^\omega - \text{Harvest}^\omega \right. \\ \left. - \text{Transport}_{H \rightarrow D}^\omega - \text{Transport}_{D \rightarrow R}^\omega - \text{OPEX}^{D,\omega} - \text{OPEX}^{R,\omega} \right. \\ \left. - \text{Slack penalty}^\omega - p_C \cdot E(\mathbf{x}^\omega, \mathbf{z}^\omega, y; \omega) \right] - \text{CAPEX}^D - \text{CAPEX}^R,$$

where OPEX denotes operating expenditure and CAPEX denotes capital expenditure. CAPEX terms are first-stage (no ω index, just y -linear); all others scenario-indexed.

3.4. County-level aggregation

The Stage-2 operational solve runs at the county level. Hex quantities aggregate to counties via area-share weighting; an invariance result for the harvest-cost term (Proposition 3, proved in Section B) supports the aggregation, with the remaining objective terms aggregating by summation or by the distance-averaging linearization documented there. The operative bound is empirical: county-level total NPV cost matches the hex-level value to within $\pm 0.15\%$ (Table 7).

3.5. Endogenous-demand variant for the regret analysis

The fixed-demand MILP above prices output at the exogenous $p_{\text{SAF}}(\omega)$ against a hard demand floor $\bar{Q}$ with slack imports. The per-scenario regret statistics of Section 5.2 and the candidate risk evaluation of Section 5.8 are computed on an *endogenous-demand* variant of the same MILP, in which the demand floor and the slack term are removed and demand clears along a linear inverse-demand curve. The variant differs from the fixed-demand model in three respects: demand and calibration, the scenario mapping, and the objective. Two further notes below are bookkeeping rather than modifications: how each scenario’s value is solved and certified, and which results use which model variant.

Demand and calibration. Inverse demand is $P^d(Q; \omega) = a_\omega - bQ$, with slope pinned by an elasticity choice at an anchor point via $b = -P^*/(\varepsilon Q^*)$ and $a = P^* + bQ^*$. The regret evaluation reported in Sections 5.2 and 5.8 is calibrated at $\varepsilon = -0.5$ (the midpoint of the $\{-0.3, -0.5, -1.0\}$ grid of Section 4.4) and the raw-mean anchor $(Q^*, P^*) = (3.0 \text{ B gal/yr}, \$5.97/\text{gal})$, giving $a = \$17.91/\text{gal}$. The regret analysis is carried at the $\varepsilon = -0.5$ calibration, not the grid’s elastic-side anchor $\varepsilon = -1.0$, because at $\varepsilon = -1.0$ the low-price wait-and-see MILPs do not converge within the time budget, so we cannot treat their recorded regret as evidence either way (Section C documents the residual gaps); converged low-price re-solves at that calibration are the computational frontier flagged in the Limitations (Section 6.4). Because the two calibrations differ in welfare scale, the *level* of $\mathbb{E}[L]$ is calibration-specific. We conjecture, but cannot verify here, that the tail-to-mean *ratio* (3.1 in the certified evaluation) is less calibration-sensitive than the level, because both the wait-and-see and here-and-now values scale with the welfare level at a given elasticity; we flag this as an open robustness item rather than an established property. The first-stage topology is invariant across the elasticity grid (Section 4.4).

Scenario mapping. Price scenarios enter as parallel intercept shifts, $a_\omega = a + (p_{\text{SAF}}(\omega) - \$5.977)$, where $\$5.977$ is the raw sample mean (quoted as $\$5.97$ elsewhere in the paper), and the $\$0.03$ raw-vs-binned gap ($\$5.977$ vs. the binned $\$6.00$) shifts all intercepts equally and cancels from every L_s ; b is held fixed, so a scenario shifts the buyer-side willingness to pay one-for-one with the wholesale price while preserving the local elasticity at the shifted anchor.

Objective. The revenue term and slack penalty in (25) are replaced by the gross-benefit integral $\int_0^Q P^d(q; \omega) dq = a_\omega Q - \frac{b}{2} Q^2$, linearized by piecewise-linear interpolation with type-2 special-ordered sets (SOS2) on a shared 21-breakpoint grid so the program remains a MILP (cross-plan comparisons are re-evaluated at a refined 81-breakpoint grid; Section 5.8); maximizing the resulting objective recovers CS + PS.

Solves. Each scenario’s wait-and-see value W_ω^{WS} is a full MILP solve at a 5% target gap, followed by a *certification* pass: a candidate plan’s fixed-siting evaluation is a feasible point of that scenario’s wait-and-see problem, so any candidate value above the recorded incumbent proves the incumbent suboptimal and triggers a re-solve warm-started from the certifying topology. The recorded W_ω^{WS} therefore dominate the full candidate

pool, and the zero-flooring of per-scenario losses is inert; the here-and-now value W_{ω}^{HN} fixes the first-stage sitings and solves the recourse problem to numerical optimality. Regret remains a lower bound on true regret up to the residual gaps; solve limits, realized gaps, and the evaluation history are reported in Section C.

Variant attribution. The deterministic baseline of Section 5.1, the information-value accounting of Section 5.3 (computed on the present-value chain surplus, PVC), and the topology-response sweep of Figure 8 use the fixed-demand model; the regret statistics of Section 5.2 and the candidate risk evaluation of Section 5.8 use the endogenous-demand variant; the closed-form objects of Section 2.4 are calibrated to the KDE alone and are variant-independent.

4. Data and Calibration

4.1. Spatial frame

The underlying spatial frame is the 14,040 50-sq-mile hex tessellation covering the 12 core SE-U.S. study states (Texas through Maryland) adopted from Li et al. (2026). We aggregate hex-level quantities to 1,374 county-level supply nodes via U.S. Census Bureau (2023) county polygons; of the 14,040 hexes, 8,864 (63%) straddle multiple counties and are assigned to counties by area-share weighting (Section B). The invariance result of Proposition 3 establishes that the harvest-cost term aggregates without distortion under its stated hypotheses; the full accounting stack matches across resolutions to within $\pm 0.15\%$ empirically (Table 7), so the empirical results of Section 5 are read at the county level with that residual in mind.

On that frame, the Stage-2 instance draws residue from the 220 counties with positive Year-0 hardwood availability and sites depots and biorefineries on 483 candidate counties each — those containing at least one of the 528 industrially zoned hexes — with both catchment legs capped at 100 miles. It is solved over a ten-year horizon against the 20 price scenarios of Section 4.3. Table 8 in Section B collects the full set of instance dimensions and Table 11 in Section D the solver profile.

4.2. Residue and demand

Annual residue availability $\bar{R}_h$ is the Stage-1 LP solution under mill demand projections from USDA Forest Service (2023a), with willingness-to-harvest $W2HF = 0.28$ (Butler et al. 2016), commercial-private share $\text{ComR} = 0.338$, residue:wood ratios computed from county-level USDA Forest Service Timber Products Output (TPO) removals data (USDA Forest Service 2023b) — the logging-residue share of total removals — assembled at hex resolution as in Li et al. (2026).

The Stage-1 hardwood residue pool delivers 72.5 M dt/yr at Year 0 across 220 supply counties; the deterministic baseline of Section 5.1 draws 56.7 M dt/yr of it (78% utilization). The pool is harvest-driven — mill-demand-determined removals times county residue:wood ratios — scaled by a 0.70 retrieval cap, the same maximum-retrieval constraint the Billion-Ton assessments impose on logging residues (retrieval cannot exceed 70%, with a minimum 30% of biomass retained on site for soil productivity and more on steep slopes; Langholtz et al. 2024, Ch. 4, and U.S. Department of Energy 2016). The willingness-to-harvest fraction ($W2HF = 0.28$) and the commercial-private share ($\text{ComR} = 0.338$) enter as an inventory-availability cap; at the demand-driven harvest level that cap is slack and does not bind the pool. The pool is a *technical-availability* envelope and is not comparable to the 2023 Billion-Ton Report’s headline figures, which report production capacity under market

scenarios rather than availability: BT23’s near-term national estimate across forest resource types is 30.3 M dt/yr (Langholtz et al. 2024, Table 4.1), with roughly 19 M dt/yr of national logging residues available at stumpage prices up to \$40/dt in the mature-market scenario (Ch. 4). Our envelope is larger by construction — TPO-based residue generation net of the retrieval cap, before any market constraint — and the binding economics in the model come from harvest, transport, and processing costs rather than from the envelope itself; the baseline uses 78% of it. The demand floor is set at $\bar{Q} = 3.0$ B gal/yr, the 2030 interim target of the U.S. SAF Grand Challenge (U.S. Department of Energy and U.S. Department of Transportation and U.S. Department of Agriculture 2021). We read it as a long-run equilibrium volume for the chain’s operating window, not a near-term forecast against 2025 actuals of $\sim 0.5\text{--}0.7$ B gal/yr; all instrument outlays in Section 5.6 are quoted per this volume and scale linearly in it. The baseline specification is hardwood-only; a softwood-inclusive all-residue sensitivity (pool 117.0 M dt/yr) re-solves the full nine-cell $\varepsilon \times \text{OPEX}$ grid and three price anchors and returns the 248/34 first-stage topology in every cell (realized gaps 3.3–14.3%; Section 6) — σ^* itself is invariant to the feedstock scope by construction, so the topology is the object the sensitivity tests. Catchment-radius construction, the sustainable-removal fraction, and the hex-county aggregation are documented in Section B.

4.3. SAF price uncertainty

Because a deep SAF spot market does not yet exist, the uncertain revenue price is a *constructed drop-in biofuel price*: a slate-weighted wholesale gasoline–diesel price adjusted to a common real basis, plus the market value of the Renewable Fuel Standard (RFS) compliance credits the fuel generates — the weekly D-code Renewable Identification Number (RIN) price scaled by the 1.7 cellulosic equivalence factor. The series spans August 2013 through December 2021 at weekly source resolution and is the producer-revenue price of the prior FP/HT-pathway calibration (Li et al. 2026), carried over as the drop-in fuel price; its construction and sources are documented in Section C. Its policy content is handled explicitly rather than assumed away. The *level* embeds RIN value (on average \$2.96/gal of the \$5.98/gal mean), so the price is policy-inclusive and the non-capitalization hypothesis of Proposition 1 must be re-argued (Section 5.5). The *dispersion*, by contrast, is genuine revenue risk to the producer — RIN-price volatility passes one-for-one into the margin of a producer paid partly in credits — and the dispersion, not the level, is what the risk objects of Section 2.4 price. The outside-option price $p^{\text{jet}} = \$3.50/\text{gal}$ (Section 2.2) is a fossil-jet benchmark on a policy-free basis; the two prices are deliberately on different bases, and the wedge between them is predominantly the embedded RIN component. We discretize the distribution with a Gaussian KDE on 20 equal-width bins (Table 2). The raw sample mean is \$5.97/gal; the probability-weighted mean of the 20-bin discretization is $\overline{p_{\text{SAF}}} = \$6.00/\text{gal}$ on support [$\$3.81, \9.12]/gal (bin-conditional means span \$3.99–\$8.94/gal). Quantities derived from the discretized distribution — the two-state collapse of Section 2.4 and the instrument calibration of Section 5.6 — use the binned mean \$6.00; the deterministic baseline of Section 5.1 is the central-scenario solve at bin 9’s conditional mean \$6.05, the bin containing the raw mean (a companion re-solve at \$5.97 itself sits on the topology margin; see the source-run footnote in Section 5.1).

4.4. Cost parameters

Feedstock quantities and residue-harvest costs follow U.S. Department of Energy (2016), with year-0 representative values discounted at $\rho = 0.10$. All cost numbers are in 2022 USD; the CPI factor 1.223 (2016→2022) is applied to the original 2016 residue-harvest-cost (RHC) values.

Table 2: KDE-discretized SAF-price scenarios

ω	Left edge (\$/gal)	Right edge (\$/gal)	p^ω	$p_{\text{SAF}}(\omega)$ (\$/gal)	Cum. p
1	3.813	4.078	0.0328	3.986	0.0328
2	4.078	4.343	0.0593	4.224	0.0920
3	4.343	4.609	0.0750	4.468	0.1671
4	4.609	4.874	0.0749	4.727	0.2420
5	4.874	5.139	0.0675	4.999	0.3094
6	5.139	5.404	0.0671	5.284	0.3765
7	5.404	5.669	0.0743	5.532	0.4508
8	5.669	5.934	0.0772	5.821	0.5280
9	5.934	6.200	0.0635	6.049	0.5915
10	6.200	6.465	0.0444	6.344	0.6360
11	6.465	6.730	0.0395	6.610	0.6754
12	6.730	6.995	0.0481	6.877	0.7235
13	6.995	7.260	0.0581	7.114	0.7816
14	7.260	7.525	0.0607	7.400	0.8423
15	7.525	7.791	0.0565	7.655	0.8988
16	7.791	8.056	0.0454	7.919	0.9442
17	8.056	8.321	0.0289	8.168	0.9731
18	8.321	8.586	0.0148	8.451	0.9880
19	8.586	8.851	0.0079	8.697	0.9958
20	8.851	9.116	0.0042	8.944	1.0000
<i>Mean (probability-weighted)</i>			1.0000	6.001	—

Notes: Twenty equal-width bins fit on EIA 2014–2024 monthly wholesale Gulf-Coast jet-A spot data with Silverman bandwidth. Bin probabilities p^ω sum to 1; conditional means $p_{\text{SAF}}(\omega)$ are bin-conditional expectations. Distribution mean \$6.00/gal, support [\$3.81, \$9.12]/gal.

Conversion-side capacities and unit costs come from two techno-economic models rather than from the feedstock literature. The biorefinery basis is a gasification–Fischer–Tropsch TEA in the lineage of [Tanzil et al. \(2021\)](#): total capital investment is built up from delivered equipment cost using the direct-cost and fixed-capital ratio factors of [Peters, Timmerhaus, and West \(2004\)](#), on the gasification reference design, operating-hour basis, and process flows of [Swanson et al. \(2010\)](#), escalated to the analysis year with the Chemical Engineering Plant Cost Index. That model is configured to this study’s feedstock and product slate, so the capital figure entering the optimization is one of its outputs rather than a number read from a published design case: the output is placed on a 100 M gal/yr anchor, scaled to other capacities with a 0.7 power law, taken net of a 10% salvage value in year 20, and annualized over a 20-year life at $\rho = 0.10$. This yields \$947 M of per-facility capital and an annualized charge of \$111.3 M/yr (Table 3). The depot basis is this project’s own preprocessing techno-economic model, in which capital and operating costs are resolved separately for the hammer-mill and crumbler size-reduction

configurations; the crumbler configuration is the one carried into the results below.

Comparison against public design cases. The charge is fixed per facility, so its per-gallon reading falls as realized output rises: the \$947 M spans \$9.5–\$19/gal-yr across the admissible [50, 100] M gal/yr band, and the baseline footprint operates near the upper end, where the implied figure is $\approx$ \$11/gal-yr. Table 3 places that figure against the two public National Renewable Energy Laboratory design cases: the thermochemical indirect-liquefaction design implies \$9.31/gal-yr of operating output on an n th-plant basis (Tan et al. 2015, 2017), and the 2018 biochemical design cases \$22–23 per annual GGE (Davis et al. 2018), so the hydrocarbon-biofuel design range spans roughly \$9–23. Our \$11 is a $\approx 1.2\times$ premium over the thermochemical n th-plant anchor — conservative relative to the $\approx 2\times$ pioneer-plant cost growth documented by Merrow, Phillips, and Myers (1981) for first-of-a-kind process plants. Neither design case enters the optimization: the comparison is a plausibility check on the derived figure, not a target.

Operating cost is \$1.96/gal at the baseline, the same TEA’s operating-cost output for this feedstock and product slate: gross operating cost per dry tonne of feedstock, less the credit for the diesel, gasoline, and naphtha co-produced alongside the jet fraction, converted to a per-gallon basis at the pathway’s liquid yield. The figure is therefore net of co-product revenue, which is why it sits well below the gross conversion cost. A further pairing deserves flagging rather than defense: the operative dry-ash-free yield of 92.84 gal/dt (Section 4.4, BCY paragraph) sits at the hydrogen-boosted end of gasification–FT performance, and if achieving it requires purchased hydrogen, \$1.96 understates operating cost — the high-yield/low-OPEX corner is an optimistic baseline. The nine-cell sensitivity over price elasticity $\varepsilon \in \{-0.3, -0.5, -1.0\}$ and OPEX $\in \{1.96, 2.30, 2.70\}$ /gal brackets exactly that possibility: the upper OPEX cells price an H₂-inclusive configuration.² Finally, we state what the TEA parameters do and do not bind. They do not bind the headline σ^* , which depends only on the risk-aversion weight and the KDE primitives and is numerically invariant across the grid by construction (Theorem 1). They do bind the build-vs-slack crossing and the regret levels. Re-solving the crossing region at each operating cost puts the crossing at \$5.49–\$5.54/gal at the \$1.96 baseline, \$5.91–\$6.06 at \$2.30, and \$6.31–\$6.44 at \$2.70 (Table 12): each per-gallon dollar of operating cost moves the crossing by slightly more than a dollar, between $1.1\times$ and $1.2\times$ on the two steps taken together. The crossing therefore rises faster than the operating-cost assumption itself, and the number of low-price bins that fall below the build threshold — the bins that carry essentially all of the regret — grows accordingly.

Biofuel conversion yield (BCY). Following the ash-tier discretization method of Li et al. (2026), BCY is discretized into 10 percentile-based tiers spanning 63.8–95.6 gal/dt across the hardwood-residue ash distribution, anchored to a base yield of 75.16 gal/dt (ash-inclusive basis) for the median-ash tier consistent with FT-from-residue performance in Tan et al. (2017); the model’s operative constant is the dry-ash-free equivalent, $75.16/0.810 = 92.84$ gal/dt, to which the per-tier ash scale factors apply. The FT pathway used here has yield characteristics distinct from the fast-pyrolysis / hydrotreatment (FP/HT) pathway analyzed in Li et al. (2026); we adopt the ash-tier *discretization scheme* from that work while substituting the FT-base BCY of ~ 75 gal/dt.

²The grid is reported in Table 12. The 248-depot / 34-refinery first-stage topology holds in eight of the nine cells; the exception is the elastic, highest-cost corner ($\varepsilon = -1.0$, \$2.70/gal), which returns 185/27 at a 10.6% realized gap and so is an unconverged incumbent rather than a certified topology shift, though its direction — a smaller footprint under elastic demand and high conversion cost — is the economically expected one. The softwood-inclusive re-solve of the same grid (Section 4) returns 248/34 in all nine cells.

Table 3: Biorefinery capital and operating cost against the public NREL design cases

Study	Pathway	Basis (cost yr)	TCI (\$M)	Output (M gal/yr)	TCI/gal-yr (\$)	Feedst. (\$/dt)
This study	residue FT	FOAK (2022)	947 ^a	50–100	9.5–19 ^a	endog. ^b
Tan et al. (2015)	IDL gasoline	<i>n</i> th (2011)	437.5	47.0	9.31 ^c	80.00
Davis et al. (2018), BDO	biochem.+adipic	<i>n</i> th (2016)	696.8	31.3 ^d	22.26 ^d	71.26 ^e
Davis et al. (2018), acids	biochem.+adipic	<i>n</i> th (2016)	758.4	32.5 ^d	23.34 ^d	71.26 ^e

Notes: Cost years differ across rows and are not deflated to a common basis; the comparison is of order, not of cents. Merrow, Phillips, and Myers (1981) document pioneer-process-plant costs roughly doubling relative to design estimates, supporting the first-of-a-kind (FOAK) multiplier of 1.5–2.5 $\times$ over *n*th-plant totals commonly applied in bioenergy TEA; our \$11/gal-yr baseline is a $\approx 1.2\times$ premium over the thermochemical *n*th-plant anchor and conservative against that range.

^a Implied per-facility total at the 100 M gal/yr capacity bound (annualized charge \$111.3 M/yr, 20-yr annuity at $\rho = 0.10$); the fixed per-facility charge over the admissible 50–100 M gal/yr band spans \$9.5–\$19/gal-yr, with the baseline solution operating near the upper bound ($\approx \$11$).

^b Stumpage, harvest, and logistics are endogenous model outcomes rather than a delivered-price assumption.

^c Per operating gallon (90% on-stream); the report’s \$8.4/gal-yr figure is per nameplate gallon.

^d Million GGE per year and \$ per annual GGE.

^e INL 2030 target delivered price, not an observed market price.

5. Results

This section reports five findings. *Result 1*: price uncertainty matters, but its welfare cost is concentrated in the low-price tail. Expected regret, the value of perfect information for a fixed plan, is \$299 M/yr, while $CVaR_{0.95}[L]$ is \$921 M/yr (3.1 $\times$ at the recorded solves; the residual-gap band on the ratio is [1.1, 4.1]), on a certified evaluation (Section 5.2). *Result 2*: a fixed-demand chain-surplus analog of the same information question gives a candidate-based EVPI estimate of \$255 M/yr (bias direction undetermined; Section D).

Result 3: the risk-premium component of the capacity-restoring credit is $\sigma^* = \$0.52/\text{gal}$ at $\lambda = 0.5$, materially below the statutory \$1.30–\$1.45/gal Section 45Z credit. Price risk alone does not cover the statutory credit at *any* risk weight: $\sigma^*(\lambda) \leq \$1.04/\text{gal}$ on all of $\lambda \in [0, 1]$. *Result 4*: adding the Pigouvian component via Proposition 1, the statutory credit is consistent with carbon plus risk at an implied social cost of carbon (SCC) of \$77–\$91/tCO₂ under the baseline reading, between the two federal SCC vintages of \$51 (Interagency Working Group on Social Cost of Greenhouse Gases 2021) and \$190 (U.S. Environmental Protection Agency 2023); Table 6 reports the full schedule of readings, including the post-2025 in-force rate at \$22–\$30/tCO₂.

Result 5: delivering the same risk correction through a one-sided price floor instead of a flat credit cuts its expected fiscal outlay by 31% (\$0.48 B/yr at the 3.0 B gal/yr anchor volume), and a two-sided contract-for-difference delivers it at zero expected outlay (Proposition 2).

The deterministic baseline, the central-scenario solve at bin 9’s conditional mean \$6.05/gal (the bin containing the raw mean \$5.97; Section 4), places a 248-depot / 34-refinery topology delivering 3.32 B gal/yr of domestic SAF at a chain surplus of \$1.43 B/yr, and provides the reference siting from which the results are computed. Instrument outlays in Result 5 are quoted at the 3.0 B gal/yr Grand Challenge anchor volume, so they are invariant to which deterministic run defines the baseline. Units convention: unless a table row is explicitly labeled otherwise, all reported quantities, including EVPI, are single-year-equivalent flows (\$/yr in year-0 terms). Operations are evaluated over a 10-year decision horizon; capital is annuitized over a 20-year asset life at $\rho = 0.10$. Result 2’s

Table 4: State-level distribution of placed depots and refineries at the deterministic baseline (KDE-mean scenario, $p_{\text{SAF}} = \$5.97/\text{gal}$)

State	Depots	Share (%)	Refineries	Share (%)
Tennessee (TN)	52	21.0	6	17.6
Virginia (VA)	46	18.5	7	20.6
North Carolina (NC)	34	13.7	4	11.8
Georgia (GA)	32	12.9	4	11.8
Alabama (AL)	26	10.5	4	11.8
Mississippi (MS)	20	8.1	2	5.9
South Carolina (SC)	13	5.2	3	8.8
Arkansas (AR)	13	5.2	2	5.9
Kentucky (KY)	8	3.2	1	2.9
Louisiana (LA)	4	1.6	1	2.9
Total	248	100.0	34	100.0

Notes: The SE-U.S. core (TN, VA, NC, GA, AL) hosts 77% of all depots and 74% of all refineries; remaining placements track the residue resource base across Mississippi, Arkansas, South Carolina, Kentucky, and Louisiana.

information-value accounting is computed on the chain-surplus (PVC) objective of the fixed-demand model, Result 1’s regret statistics on the endogenous-demand variant (Section 3.5); the two live on different objectives and are not compared to each other.

5.1. Deterministic baseline

We first solve the deterministic single-scenario problem at the central scenario — bin 9 of Table 2, conditional mean $p_{\text{SAF}} = \$6.05/\text{gal}$, the bin containing the raw sample mean $\$5.97$ (Section 4). The incumbent facility configuration places $N_D = 248$ depots and $N_R = 34$ biorefineries across 10 of the 12 study states, producing 3.32 B gal/yr of domestic SAF and fully meeting the 3.0 B gal demand floor (zero slack imports); present-value chain surplus is $\$1.43$ B/yr.³ The deterministic siting tracks the SE-U.S. forest-residue base (Figure 3); state-level shares and the ash-tier allocation are reported in Table 4 and Section D.

5.2. Price uncertainty and the low-price tail

The central here-and-now plan $y_{\text{bin } 9}^*$ is the central-scenario siting of Section 5.1 (bin 9 of Table 2); it serves as the here-and-now reference against which the wait-and-see ideal is benchmarked. Its welfare cost is concentrated in the *low-price* tail. Figure 4 reports the per-scenario welfare loss $L_s = W_s^{\text{WS}} - W_s^{\text{HN}}$ across the 20 KDE bins of the endogenous-demand variant (Section 3.5), where W_s^{WS} is the recorded scenario- s wait-and-see value and W_s^{HN} is the realized welfare of the here-and-now plan $y_{\text{bin } 9}^*$ evaluated at scenario s .

The certified evaluation gives one column with a transparent bias direction. The loss declines monotonically across the nine lowest-price bins ($L_s = \$947, \$872, \$781, \$671, \$554, \$432, \$325, \180 , and $\$48$ M/yr at bins

³The bin-9 plan is a feasible incumbent, not a certified optimum; Section D reports the source runs, a companion re-solve at the raw mean price, and their realized gaps. The regret analysis of Section 5.2 treats it as a fixed commitment rather than an optimality claim.

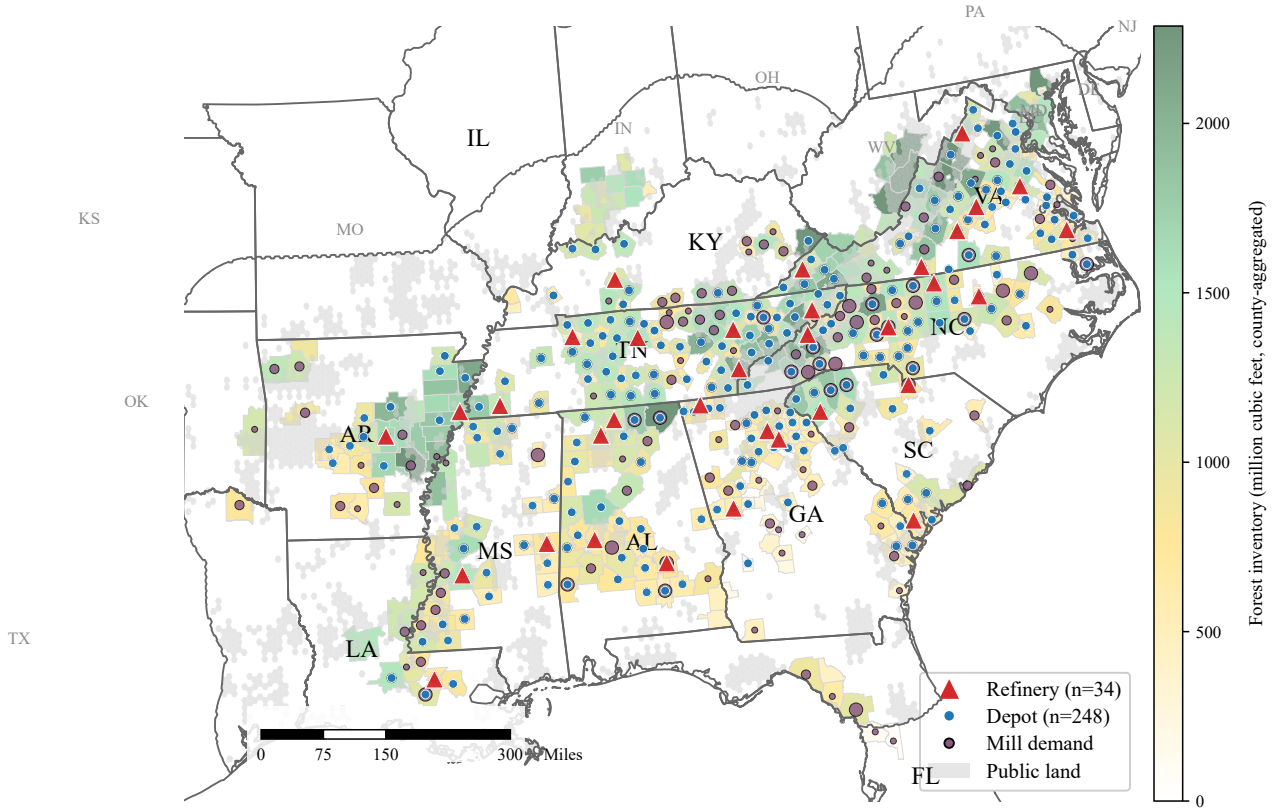


Figure 3: County-level baseline siting map: forest residue inventory (background shading) overlaid with the placed depots and refineries of the deterministic baseline plan (bin-9 central scenario, $p_{SAF} = \$6.05/\text{gal}$; 248 depots, 34 refineries).

1 through 9) and is zero in the 11 highest-price bins ($\approx 41\%$ probability mass), where the certified wait-and-see topology coincides with the committed plan. Expected regret is $\mathbb{E}[L] = \$299 \text{ M/yr}$ and the tail is $\text{CVaR}_{0.95}[L] = \921 M/yr (the worst-5% window comprises bin 1 in full plus the lowest 29.0% of bin 2's mass: $1 - \alpha = 0.05 = p^1 + 0.290 p^2$ with $p^1 = 0.0328$, $p^2 = 0.0593$ from Table 2), a tail-to-mean ratio of 3.1. Because the wait-and-see values are converged incumbents (gaps 1.3–3.9%), these figures remain lower bounds on true regret: the gap-implied allowance is $\$528 \text{ M}$ in expectation, so $\mathbb{E}[L]$ lies in $[\$299, \$827] \text{ M/yr}$, with certification pinning the floor against the entire candidate pool. One property of the certified evaluation should be stated plainly: in all twenty bins the recorded wait-and-see value coincides, to machine precision, with the best fixed-siting value in the six-plan candidate pool — the certification re-solves never improved on their warm starts. The reported regret is therefore a *best-in-pool* regret: a candidate-based lower bound on the value of perfect information with no wait-and-see content beyond the pool, and the zero losses in bins 10–20 hold by construction (the committed plan is the pool's argmax there), not because those solves closed their gaps. The allowance is documented bin by bin in Table 10; carried jointly rather than through $\mathbb{E}[L]$ alone, it bounds $\text{CVaR}_{0.95}[L]$ within $[\$921, \$1,235] \text{ M/yr}$ and the tail-to-mean ratio within $[1.1, 4.1]$. The recorded point is 3.1, but an adversarial allocation of the allowance to the zero-loss high-price bins would compress the concentration toward parity, so we report the ratio with its band rather than as a point. The earlier evaluation vintages and the certification protocol are documented in Section C.

The economics of the concentration is overcommitment, and the certified wait-and-see topologies trace it

directly. The best-known scenario-optimal siting is 192/26 depots/refineries at bin 1, 197/27 across bins 2–9, and the committed 248/34 from bin 10 upward: a weakly monotone expansion in price. In bins 1–9 the committed plan’s excess capital relative to the smaller ideals is the regret, including a \$48 M shortfall at bin 9 (the committed plan’s own calibration bin, which sits within that solve’s optimality gap). At high prices marginal capacity beyond the central scale earns little (incremental catchment residue is expensive and the consumer-surplus integral flattens), and no certified wait-and-see topology exceeds 248/34 in any bin. Commitment risk is therefore asymmetric: overbuilding against a low price is costly, while underbuilding against a high price is cheap at the margin. It is not free, though. As Section 5.8 shows, the missed upside keeps the committed plan ahead of the smaller plans on the *regret* tail even as they beat it in welfare levels. This low-price tail concentration motivates the mean–CVaR objective in (4).

The regret L_s and the CVaR functional price different objects, and they should not be conflated even though both point at the same tail. The regret L_s plotted in Figure 4 measures the cost of having committed capacity before the price was known; the CVaR functional in (4) instead prices tail risk in welfare *levels* ($-W$), and it is the welfare-level tail aversion that generates the σ^* of Theorem 1 and Result 3. Empirically the two coincide in location. The same per-scenario evaluation puts the welfare-level tail in the lowest price bins as well: $\text{CVaR}_{0.95}$ of the committed plan’s welfare level sits 33% below its expectation, carried by the two lowest-price bins, so the low-price states that carry the regret are also the states the mean–CVaR objective weights. Result 1’s regret statistics thus point at exactly the tail that Theorem 1 prices, while remaining a different object from it: regret quantifies the value of information and the cost of commitment, not the argument of the mean–CVaR functional, and we keep the symbols distinct (L_s for regret, $-W$ for welfare loss) for that reason.

The distinction has bite. Section 5.8 shows the two objects rank *plans* in opposite directions: the committed plan minimizes the regret tail within the candidate set, while a smaller plan dominates it in welfare levels at every risk weight.

5.3. The information question on the chain-surplus accounting

Result 1’s expected regret is, for the fixed central plan, the value of perfect information computed on the endogenous-demand welfare objective. Because the committed plan is fixed rather than recourse-optimal, that \$299 M/yr is an upper bound on the endogenous-demand EVPI of (5) relative to the candidate pool; on the refined evaluation grid the committed plan is itself the pool’s expected-welfare argmax (Section 5.8), so $\min_c \mathbb{E}[L \mid y_c]$ coincides with the committed plan’s own \$300 M/yr and the candidate-based estimate collapses onto the headline figure. The wait-and-see incumbency gap still pushes in the opposite direction, so, like the \$255 M figure below, this is not a certified bound. This subsection asks the same information question on the fixed-demand chain-surplus (PVC) accounting, a robustness check at a different welfare scale, not a comparable number (Section 3.5). Two honesty notes govern the accounting. First, the “recourse-problem” entry in Table 5 is not a solved two-stage MILP: it is $\max_c \mathbb{E}[W \mid y_c]$ over the twenty wait-and-see candidate plans (the maximizer is the bin-9 plan), which is a valid lower bound on the true recourse optimum $W^{\text{RP}*}$; the extensive form was not solved. Second, the wait-and-see expectation is incumbent-weighted ($\sum_{\omega} p^{\omega} \text{LB}_{\omega}$), a lower bound on the true $\mathbb{E}[W^*]$. The difference of two lower bounds has no determinate bias sign, so the \$255 M/yr figure is a *candidate-based estimate* of EVPI rather than a certified bound; Section D details both bias channels.

The best-candidate (RP-proxy) siting is a strict subset of the union of the 20 wait-and-see (WS) per-scenario

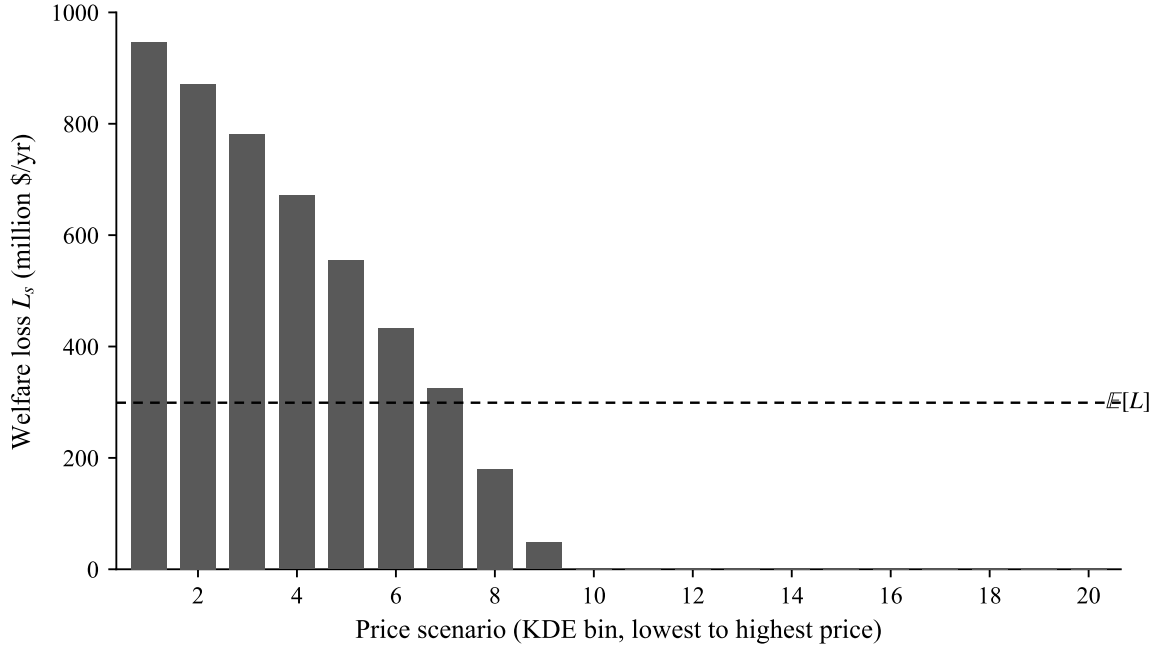


Figure 4: Per-scenario welfare loss $L_s = W_s^{\text{WS}} - W_s^{\text{HN}}$ of the committed here-and-now plan $y_{\text{bin } 9}^*$ across the 20 KDE-discretized SAF-price scenarios (endogenous-demand variant, Section 3.5; certified evaluation: every wait-and-see value dominates the full candidate pool, realized gaps 1.3–3.9%). The loss declines monotonically from \$947 M/yr at bin 1 to \$48 M/yr at bin 9 — where the wait-and-see ideal is a structurally smaller chain (192/26 at bin 1, 197/27 at bins 2–9) — and is zero from bin 10 upward. The dashed line marks $\mathbb{E}[L] = \$299$ M/yr; $\text{CVaR}_{0.95}[L] = \921 M/yr, carried by the two lowest-price bins. Losses remain lower bounds on true regret up to the residual gaps.

optima. All 248 depots placed by the RP-proxy also appear in at least one WS solution (100% overlap); zero are RP-exclusive. The WS union covers 330 distinct candidate locations, of which 82 are WS-only, depots that specific scenarios would prefer to build but that the recourse policy shared across all 20 scenarios elects to forgo. Figure 6 (in Section F) panel (a) shows the scenario-recurrence distribution: RP-included depots appear in many WS solutions (mode at $\geq 18/20$), while WS-only depots typically appear in fewer than 5 scenarios. This is consistent with the interpretation that the recourse policy “concentrates” the chain by picking depot/refinery locations that are robust across the scenario set, deferring scenario-specific marginal capacity to be handled by routing flexibility rather than by siting commitment. Panel (b) shows the per-state breakdown: the overlap is preserved across the Southeast core (TN, VA, NC, GA, AL), with the WS-only tail concentrated in the lower-feedstock-density states where individual scenarios prefer geographic diversification that the recourse policy does not value.

5.4. A welfare-equivalent subsidy below the 45Z credit

The welfare-equivalent subsidy $\sigma^* = \$0.52/\text{gal}$ is the per-gallon transfer that restores risk-neutral capacity for a mean-CVaR planner facing price risk and irreversible investment. It prices the same low-price welfare tail that carries the regret of Result 1, though it is calibrated to the KDE primitives directly, not to either MILP variant’s statistics, and it is the policy-relevant number that we read against the statutory \$1.30–\$1.45/gal Section 45Z credit.

Table 5: Information-value accounting on the chain-surplus (PVC) objective at the deterministic 248-depot/34-refinery baseline

Quantity	Value (\$ million)
$\sum_{\omega} p^{\omega} \text{LB}_{\omega}$, incumbent-weighted WS expectation (lower bound on $\mathbb{E}[W^*]$)	1,541
$\max_c \mathbb{E}[W \mid y_c]$, best candidate plan (lower bound on $W^{\text{RP}*}$; extensive form not solved)	1,286
W^{EEV} value of expected-value solution	n.r. (Section D)
EVPI , candidate-based estimate (difference of the two rows above)	255 (19.8%)
VSS ($= W^{\text{RP}} - W^{\text{EEV}}$)	n.r. (Section D)

Notes: All quantities are single-year-equivalent surpluses (\$M/yr; capital as a 20-year annuity at $\rho = 0.10$), computed on the fixed-demand model (Section 3.5, variant attribution); the EVPI percentage is relative to the best-candidate RP-proxy row (255/1,286); they are not commensurable with the endogenous-demand regret statistics of Section 5.2. Both reported rows are lower bounds on their true objects (the WS expectation is incumbent-weighted; the RP row is a best-candidate evaluation in the spirit of the candidate-evaluation literature, [Mak, Morton, and Wood 1999](#); [Bayraksan and Morton 2011](#), though we do not run their replication schemes), so their difference is a candidate-based EVPI estimate with undetermined bias sign, not a certified bound (see Section D). EEV-based rows and VSS are not reported (n.r.) because they would require a further nonzero-gap MILP without an anchoring bound.

Closed-form. The collapsed-regime expression (8) of Theorem 1 delivers

$$(26) \quad \sigma^* = \lambda \pi_H (a_H - a_L),$$

which holds whenever the CVaR confidence α exceeds the upside probability π_H ; at $\alpha = 0.95$ and the KDE-implied $\pi_H = 0.472$ this is the relevant regime.

KDE-anchored calibration. The 20-bin empirical KDE on the SE-U.S. wholesale SAF price distribution (Table 2, Section 4) admits a two-state median collapse at the bin-8/9 boundary (\$5.93, the bin edge nearest the binned KDE mean $\overline{p_{\text{SAF}}} = \$6.00/\text{gal}$): probability masses $\pi_H = 0.472$ above and $\pi_L = 0.528$ below, with conditional means $a_H = \$7.16/\text{gal}$ and $a_L = \$4.97/\text{gal}$, so the state-conditional demand-intercept spread is $a_H - a_L = \$2.19/\text{gal}$. The two-state primitives are computed from the KDE scenario table alone; the regret statistics of Section 5.2 use the same 20-bin scenario set but a specific demand calibration (Section 3.5), so the two share the price distribution, not a model.

Risk-aversion weight λ . We report σ^* as a schedule in λ rather than defend a point. The closed form is linear at fixed KDE primitives, $\sigma^*(\lambda) = \lambda \times \$1.04/\text{gal}$, so the admissible range $\lambda \in [0, 1]$ maps to $\sigma^* \in [0, \$1.04]/\text{gal}$. The endpoint $\lambda = 0$ recovers the risk-neutral planner of [Weitzman \(1974\)](#), which by Corollary 2 gives $\sigma^* = 0$; the endpoint $\lambda = 1$ recovers the tail-only specification, whose closed-form optimum — in the collapsed regime $\alpha \geq \pi_H$, which includes the baseline — is the deterministic optimum at the low-state conditional mean, and whose nearest empirical counterpart, the converged bin-1 wait-and-see plan, sites 192 depots / 26 refineries (Section 5.8). One statement is free of the choice entirely: since $\sigma^*(\lambda) \leq \$1.04 < \$1.30/\text{gal}$, *price risk alone cannot cover the statutory credit at any risk weight* — the level result holds for every $\lambda \in [0, 1]$, not only at a calibrated point. Where a point is needed for illustration we use the midpoint $\lambda = 0.5$ and carry λ explicitly in every downstream object; an external anchor from observed project-finance hurdle premia (inverting the closed

form through the capital stack, $\lambda = \Delta\rho \times 11/1.04$ at the \$11/gal-yr TCI calibration) belongs to the companion paper’s project-finance setting (Li and Yu 2026).

Headline number and policy reading. At $(\lambda, \pi_H, a_H - a_L) = (0.5, 0.472, \$2.19)$, equation (26) delivers $\sigma_{\text{KDE}}^* = 0.5 \times 0.472 \times \$2.19 \approx \$0.52/\text{gal}$ as the welfare-frame, per-unit risk-premium subsidy at the SE-U.S. pioneer-plant calibration. The number admits two readings. First, $\sigma^* = \$0.52/\text{gal}$ is the planner-side benchmark that closes the welfare-frame output gap; it is materially below the statutory \$1.30–\$1.45/gal Section 45Z credit, so price risk alone rationalizes only part of the observed subsidy — and the shortfall is not an artifact of the midpoint calibration, since even $\lambda = 1$ delivers only \$1.04/gal. How much of the remainder reflects the carbon externality the statute is scaled to? The decomposition of Section 5.5 answers that question. Second, σ^* is a *lower bound* in a specific, closed-form-disciplined sense: the expression $\lambda \pi_H (a_H - a_L)$ contains neither marginal cost nor curvature, so cost-level feedbacks leave it unchanged; what raises it are the channels that act on the committing agent’s effective risk aversion (financing frictions and hurdle-rate wedges; Section 6.1). Three channels work in the opposite direction: fixed-price offtake coverage scales σ^* down by the covered share; positive co-movement of input costs with fuel prices compresses the spread of the *net* premium $A = a - c$ that the closed form actually prices (a natural hedge); and partial financial hedging of the residual exposure does the same (Section 6.1). The decomposition below is conservative on the risk side only in the first sense, and both signs are carried through the discussion in Section 6.1.

5.5. Decomposing the 45Z credit: carbon plus risk

Section 45Z is by statute a carbon-intensity-scaled production credit, so its natural welfare accounting has two components: a Pigouvian payment for avoided emissions and a compensation for price risk under irreversible investment.⁴ Proposition 1 establishes that in the mean–CVaR framework the two components are exactly additive: the capacity-restoring credit is $\sigma_{\text{tot}}^* = \nu + \sigma^*$, with no interaction term. The Pigouvian component can therefore be priced without re-solving the stochastic program.

The Pigouvian component. Under the make-or-buy structure of Section 2.2, every marginal domestic gallon displaces one imported fossil gallon, so the externality value per gallon is the life-cycle emissions differential times the social cost of carbon (SCC): $\nu = \Delta e \cdot p_C$ with $\Delta e = 11.7 - 1.5 = 10.2 \text{ kgCO}_2\text{e/gal}$ on a consistent life-cycle basis (Section 2.2: CORSIA fossil baseline 89 gCO₂e/MJ at 131.5 MJ/gal against GREET-type residue-FT SAF; the CORSIA default core-LCA value for the pathway would give $\Delta e = 10.6$; certified values under 45ZCF-GREET or the provisional-emissions-rate route could differ, Section E). The two federal SCC vintages bracket the price: at the Interagency Working Group (IWG) interim value of \$51/tCO₂ (Interagency Working Group on Social Cost of Greenhouse Gases 2021), $\nu = \$0.52/\text{gal}$; at the U.S. Environmental Protection Agency value of \$190/tCO₂ (U.S. Environmental Protection Agency 2023), $\nu = \$1.94/\text{gal}$ (the Rennert et al. 2022 estimate of \$185/tCO₂ is essentially the EPA case).

The hypothesis of Proposition 1, that ν is not already reflected in the market intercept $a(\omega)$, does *not* hold cleanly here, and we do not claim it does. The revenue price is fitted on the policy-inclusive constructed series of

⁴The 2025 budget reconciliation act extended 45Z through 2029, aligned the SAF base rate with the non-SAF rate after 2025, and restricted qualifying feedstocks to North America (immaterial here: SE-U.S. forest residue is domestic). The baseline targets the as-enacted IRA SAF vintage; the post-2025 rate is repriced explicitly in Table 6.

Table 6: Implied-SCC readings of the Section 45Z credit

Reading	Assumption changed	\$/tCO ₂
Baseline (incremental)	as-enacted rate; flat credit; $\lambda = 0.5$; physical Δe	77–91
$\Delta e \pm 20\%$	physical-differential band	64–114
Instrument-efficient	risk via floor/CfD ($= \lambda = 0$; $= \theta = 1$)	127–142
Maximal risk aversion	$\lambda = 1$	26–41
Statutory emissions basis	(50 – CI) replaces physical Δe	169–180
Post-2025 rate	\$1.00/gal base $\Rightarrow$ credit \$0.74–0.83/gal	22–30
Statutory basis + stacking	joint, with \$0.30/gal stacked value (Section 6.1)	≈ 234 –239
Full RIN capitalization	embedded \$2.96/gal read as carbon support	≈ 367 –382

Notes: Each row recomputes (27) under one named change, holding the others at baseline (unrounded $\sigma^* = \$0.518$, $\Delta e = 10.2$ kgCO₂e/gal). The envelope across admissible combinations spans roughly \$22 to \$382/tCO₂: the between-vintages property belongs to the baseline incremental reading, not to the schedule.

Section 4, whose level embeds RFS compliance value averaging \$2.96/gal (Section C). The baseline inversion below is therefore an *incremental* reading: it prices the carbon channel that Section 45Z adds on top of the existing RIN stack, holding that stack in place. To the extent RIN prices themselves capitalize carbon value rather than pure RFS compliance scarcity, part of the physical differential Δe is already paid for through the price, and the incremental reading overstates the residual carbon channel available for 45Z to price; the full-capitalization outer bound is reported in Table 6 below.

Reading the statutory credit through the decomposition. Adding the risk component $\sigma^* = \$0.52$ /gal, the two-channel total spans $\sigma_{\text{tot}}^* \in [\$1.04, \$2.46]$ /gal across the two SCC vintages, a band that contains the statutory \$1.30–\$1.45/gal credit. Inverting (10) at the baseline $\lambda = 0.5$, the statutory credit is consistent with an implied SCC of

$$(27) \quad p_C^{\text{implied}} = \frac{\text{credit} - \sigma^*}{\Delta e} \in [\$77, \$91] / \text{tCO}_2,$$

between the two federal vintages (computed at the unrounded $\sigma^* = \$0.518$). Because the implied SCC scales as $1/\Delta e$, a $\pm 20\%$ band on the life-cycle differential moves it by $-17\%/+25\%$ (to \$64–\$76 and \$96–\$114 respectively), inside the inter-vintage gap in every case.

The implied SCC is a schedule, not a number. Equation (27) is one cell of a schedule in four choices the reader can see plainly: the instrument that delivers the risk component, the statutory vintage, the emissions basis, and the capitalization treatment of the embedded RIN value. Table 6 reports the named readings side by side.

Three rows deserve emphasis. The *instrument-efficient* reading takes Proposition 2 seriously: if the risk component is delivered through the floor/CfD family this paper recommends, the flat credit is left to price carbon alone and the implied SCC is $\text{credit}/\Delta e = \127 –\$142/tCO₂ — a value that also arises independently as the $\lambda = 0$ and full-offtake ($\theta = 1$) cases, and that brackets the baseline from above. The *post-2025* reading prices the credit actually in force: the 2025 reconciliation act aligned the SAF base rate to \$1.00/gal after 2025, giving a credit of \$0.74–\$0.83/gal over the pathway CI range and an implied SCC of \$22–\$30/tCO₂, *below* the IWG vintage — the in-force credit is under-sized relative to carbon-plus-risk at either federal SCC. The *full-capitalization*

bound treats the entire embedded RIN component as carbon support already paid; it is the outer bound of the capitalization channel, not a reading we endorse. We therefore do not claim that the implied SCC lies between the federal vintages as a general matter: the claim holds for the baseline incremental reading, and Table 6 shows exactly which departures break it. On the frontier, the between-vintages region also requires $\lambda \lesssim 0.75\text{--}0.90$; above that risk weight the implied SCC falls below the IWG floor.

More generally, the locus of (λ, p_C) pairs consistent with the statutory credit, the *rationalization frontier* $0.0102 \cdot p_C + 1.036 \lambda \in [1.30, 1.45]$ plotted in Figure 5, spans the territory between the two vintages. A risk-neutral planner ($\lambda = 0$) rationalizes the credit only at $p_C \in [\$127, \$142]$, well below the EPA value (at $\$190/\text{tCO}_2$ the carbon channel alone is $\$1.94/\text{gal}$, exceeding the statutory credit and leaving it *under*-sized even before risk); the IWG value rationalizes the credit only at substantial risk aversion, $\lambda \in [0.75, 0.90]$. The sign of the question “is 45Z too generous?” therefore flips with the SCC vintage. The decomposition replaces it with a sharper one: the statutory level is consistent with carbon-plus-risk for any SCC between the two federal vintages at moderate-to-high risk aversion.

What the implied SCC is and is not. Two qualifications govern the reading of (27). First, the emissions basis. The statute’s 50 kgCO₂e/mmBTU reference is a policy parameter, not a physical fossil baseline: the CORSIA fossil-jet life-cycle value is 89 gCO₂e/MJ = 93.9 kgCO₂e/mmBTU, so the statutory formula credits only $(50 - \text{CI}) = 37\text{--}41$ kgCO₂e/mmBTU ($\approx 4.6\text{--}5.2$ kgCO₂e/gal) of avoided emissions against the physical differential of $\Delta e = 10.2$. Scaling the Pigouvian component off the statutory differential instead of the physical one raises the implied SCC to approximately $\$169\text{--}\$180/\text{tCO}_2$, approaching the EPA vintage of $\$190$. We report the physical basis in (27) because the welfare-relevant object is physical avoided emissions, and flag the statutory basis as the bound on the other side. Figure 7 (in Section F) traces the full mapping: because the statutory credit and the emissions differential both move with the pathway CI, the implied SCC is a schedule rather than a point, and the vertical gap between the physical-basis and statutory-basis curves at any CI is the emissions-basis wedge. Over the FT-residue CI range the physical schedule runs $\$77\text{--}\$88/\text{tCO}_2$ and the statutory schedule $\$169\text{--}\$180/\text{tCO}_2$; the federal vintages bracket the physical curve everywhere on the relevant range. Second, *circularity*: because the statutory level is CI-scaled, the implied SCC is not an independent revealed-preference estimate of the drafters’ carbon valuation; it is the internal-consistency price at which the statutory formula and the risk premium reconcile within one welfare accounting. We report it as a benchmark against the federal SCC vintages, a diagnostic of whether the credit’s level can be spanned by its two nameable justifications, not as an identification of policymaker preferences.

What the decomposition does not include. The decomposition deliberately excludes two channels. General-equilibrium feedbacks (Section 6.1) raise the risk component, making the decomposition conservative. The private-side participation gap, the transfer a competitive chain with a project-finance hurdle rate would require (as distinct from the planner-side welfare benchmark), is an incidence question, not a welfare justification, and is treated in the companion paper (Li and Yu 2026); adding it to (10) would double count. The Pigouvian component here is priced at the make-or-buy margin off the average life-cycle differential; an endogenous-siting marginal-abatement-cost schedule from Pigouvian re-solves of the MILP at $p_C \in \{50, 100, 200\}$ $\$/\text{tCO}_2$ remains future work (Section 7).

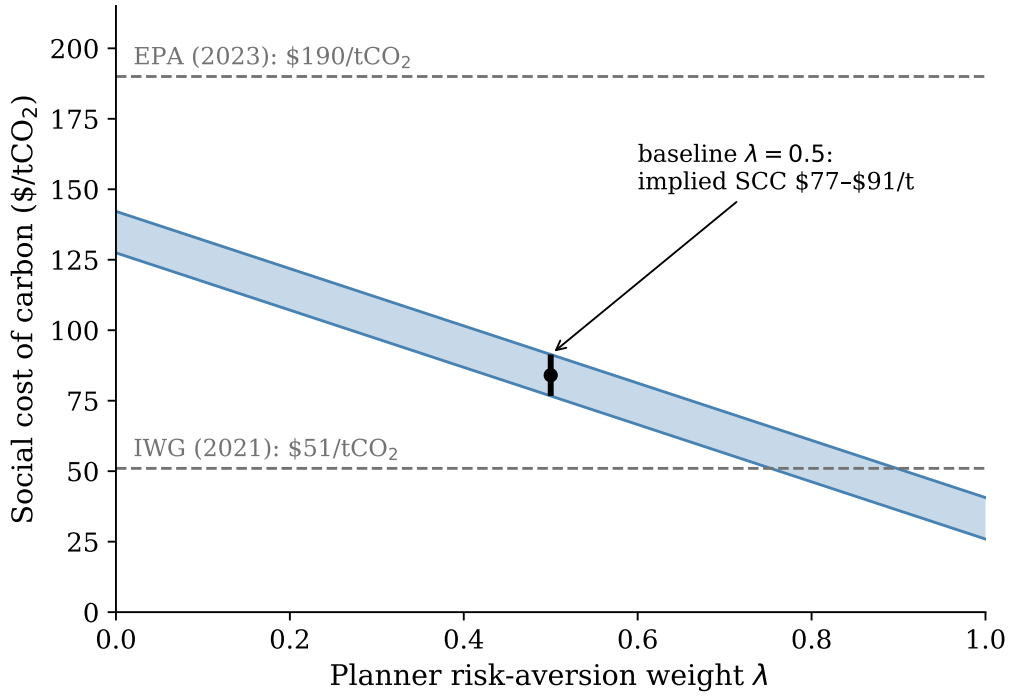


Figure 5: The 45Z rationalization frontier. Shaded band: the locus of risk-aversion weights λ and social costs of carbon p_C at which the two-channel decomposition $\sigma_{\text{tot}}^* = \Delta e \cdot p_C + \lambda \pi_H (a_H - a_L)$ of Proposition 1 exactly equals the statutory \$1.30–\$1.45/gal Section 45Z credit, at the KDE-anchored calibration ($\Delta e = 10.2 \text{ kgCO}_2\text{e/gal}$, $\pi_H(a_H - a_L) = \$1.04/\text{gal}$). Horizontal lines: the two federal SCC vintages (\$51/tCO₂, [Interagency Working Group on Social Cost of Greenhouse Gases 2021](#); \$190/tCO₂, [U.S. Environmental Protection Agency 2023](#)). Marker: the baseline $\lambda = 0.5$, at which the implied SCC is \$77–\$91/tCO₂.

5.6. Instrument form: the fiscal cost of paying for risk with a flat credit

The statutory credit pays a flat per-gallon amount in every price state. Proposition 2 shows this is the most expensive way to purchase the risk correction: a flat credit shifts the mean of the payoff distribution, while the distortion σ^* corrects is a variance phenomenon. Calibrating the proposition to the KDE primitives prices the difference.

One-sided price floor. The floor strike that restores risk-neutral capacity, (11), evaluates to $\kappa^* = \$5.64/\text{gal}$ (unrounded \$5.644) at the baseline $(\lambda, \pi_H, a_L, a_H) = (0.5, 0.472, \$4.97, \$7.16)$. (Proposition 2 writes the floor as a contract on the demand intercept $a(\omega)$; because scenarios shift the intercept one-for-one with the price (Section 3.5), the strike converts to a price floor at the same dollar value, which is how we quote it.) The strike sits six percent below the binned KDE mean of \$6.00 — a gap of exactly the size that Section 5.6’s strike-error sensitivity shows to matter. Its expected fiscal outlay (12) is $T_{\text{floor}} = \sigma^* \times \pi_L / (\pi_L + \lambda \pi_H) = \$0.52 \times 0.691 \approx \$0.36/\text{gal}$, a 31% saving on the flat credit. The two-state closed form is not driving this number: evaluating the same floor directly on the full 20-bin KDE gives $\mathbb{E}[(\kappa^* - p_{\text{SAF}})_+] = \$0.37/\text{gal}$, within 4% of the closed form. At the 3.0 B gal/yr Grand Challenge anchor volume, the flat credit’s expected outlay for the risk component is \$1.55 B/yr against \$1.07 B/yr for the floor, an expected saving of \$0.48 B/yr for the identical output correction. (Quoting outlays at the anchor volume keeps them invariant to which deterministic run defines the baseline; at the bin-9

baseline output of 3.32 B gal/yr every figure scales up by 11%.)

Two-sided CfD. A two-sided contract-for-difference at strike equal to the KDE mean (\$6.00/gal) eliminates the committing agent's price variance entirely and restores risk-neutral capacity at *zero* expected fiscal outlay (Proposition 2(iii)). The government's side of the swap is actuarially fair but not riskless: it absorbs a symmetric exposure with a per-gallon standard deviation of \$1.28 (the binned KDE dispersion).

Whose risk aversion — and why the reallocation creates value. The instrument-form result requires the two sides of the swap to price the same risk differently; Assumption 9 keeps that coherent. The tail-averse weight λ prices the exposure of the *committing* side — the Southeast regional aggregate, or equivalently the non-diversified investor whose balance sheet carries the irreversible capacity — for which the project is large and concentrated. The *federal* side pools the swapped exposure across the national taxpayer base, the Arrow and Lind (1970) condition under which its risk premium tends to zero. If instead one insisted that a single national planner both sets $\lambda = 0.5$ and holds the CfD book, the two positions would be inconsistent: Arrow–Lind applied to that planner would force $\lambda = 0$ and eliminate σ^* altogether. Keeping the claim honest requires two qualifications. First, jet-fuel prices are partially correlated with aggregate conditions, and Arrow–Lind does not cover the systematic component; to that extent the federal side bears a nonzero premium and $T_{\text{CFD}} = 0$ understates the CfD's social cost. Second, Corollary 4 maintains a state-independent marginal cost of public funds; a countercyclical $\mu(\omega)$ would add a covariance term to the CfD's deadweight cost.

Why the exposure is not simply hedged away. The obvious private substitute for the entire instrument family is a futures hedge, under which $\lambda \rightarrow 0$ and no risk instrument is warranted. The three frictions that keep the relevant exposure unhedgeable — hedging tenor, settlement basis, and margin liquidity — are the microfoundation of Assumption 9 and are stated in Section 2.4. λ prices the long-tenor, basis-laden residual those frictions leave, bounded below the value a fully unhedged reading would justify; this is the third downward channel on σ^* noted in Section 6.1.

Strike risk and settlement basis. The zero-outlay result holds at a strike equal to the true mean, which is exactly the object the thin SAF market makes hard to observe. The sensitivity is linear and transparent: a two-sided CfD at strike $\bar{a} + \delta$ has expected outlay exactly δ per gallon, so the CfD's advantage over the floor vanishes at a strike error of $\delta = \$0.36$ (6% of the mean) and its advantage over the flat credit at $\delta = \$0.52$ (9%). The one-sided floor is more robust to the same error: its outlay rises at slope $\pi_L \approx 0.53$ per dollar of strike error rather than one-for-one, which strengthens the case for the floor as the practically robust member of the family. Settlement is the second practical constraint: no deep SAF price index yet exists (Section 4), so a contract written today would settle against jet-A plus a premium, leaving the producer exposed to premium basis risk that the instrument cannot remove. Both considerations temper the two-sided CfD's headline zero and leave the floor as the implementable recommendation.

Costing the floor as an implementable contract. The floor we price in this section is implementable with existing contract technology, but not through the tax code alone. *Settlement index:* no deep SAF spot market exists — Argus and Platts publish U.S. SAF assessments (Gulf Coast, Illinois, West Coast), but the market remains

thin enough that these are indicative rather than transaction-based ([Argus Media 2025](#); [S&P Global Commodity Insights 2025](#); [Airlines for America 2025](#)) — so a workable contract would settle on a liquid Gulf-Coast jet-A assessment plus a SAF adder that is either fixed administratively or discovered at auction. *Template*: the U.K. Contracts-for-Difference scheme shows both the mechanics and the strike-setting risk. Strikes are discovered in pay-as-clear auctions capped by administrative strike prices, with a 15-year tenor, CPI indexation, and a government-owned counterparty (the Low Carbon Contracts Company) ([Low Carbon Contracts Company 2025](#)). Allocation Round 4 (2022) cleared offshore wind at £37.35/MWh (2012 prices); Round 5 (2023) set the administrative cap at £44/MWh and attracted *zero* offshore-wind bids after costs rose; Round 6 (2024) raised the cap to £73/MWh and cleared at £58.87/MWh ([Regen 2022](#); [Offshore Renewable Energy Catapult 2023](#); [UK Department for Energy Security and Net Zero 2024](#)). The lesson is that auctions reveal costs, but administered caps set on stale information can misfire in either direction — directly relevant to setting a SAF strike from TEA estimates. *Required elements*: a U.S. SAF floor would need a tenor matched to project debt (10–15 years), a per-facility volume cap, CPI indexation of the strike, competitive strike discovery with an administrative reserve, an explicit allocation of SAF-versus-jet-A basis risk (settling on jet-A leaves the SAF premium with the producer; settling on a SAF assessment imports thin-index risk), and a creditworthy counterparty. Because Treasury’s credit machinery is ill-suited to contingent two-sided payouts, natural administrators are the DOE Loan Programs Office’s project-finance apparatus or a dedicated federally backed counterparty analogous to the LCCC ([U.S. Department of Energy, Loan Programs Office 2025](#)). These frictions are not free: auction administration, counterparty capitalization, and monitoring absorb resources, and the \$0.48B/yr saving we report is gross of these administrative and transaction costs. Institutional detail on certification, rates, transferability, and stacking is collected in Section E.

Welfare stakes under costly public funds. At marginal cost of public funds $\mu = 1$ the instruments are welfare-equivalent and the comparison concerns expected outlay only (Corollary 4). With the curvature K calibrated from the MILP (Section 5.8), the comparison becomes a benefit–cost test, and it is one-sided. The deadweight loss the risk correction recovers is $\sigma^{*2}/(2K) \approx \31 M/yr (\$31–\$65 M across the K fits), while the MCPF cost of paying for it with a flat credit at the empirical federal range $\mu \in [1.10, 1.30]$ ([Dahlby 2008](#)) is \$155–\$466 M/yr: the flat credit fails benefit–cost for the risk component at any $\mu > 1.02$. The floor’s MCPF cost (\$107–\$322 M/yr) also exceeds the recovered loss except at the most favorable corner of the ranges; only the two-sided CfD, with zero expected outlay, passes unambiguously. Costly public funds therefore do more than rank the instruments: they imply the risk component should not be delivered through an outlay-bearing instrument at all — sharpening the design implication below.

Design implication. The decomposition of Section 5.5 and the ranking of Proposition 2 together imply a mixed design: the Pigouvian component is a payment for a state-independent externality and is correctly delivered flat, while the risk component is a payment for variance and is more cheaply delivered as a price floor or CfD — the instrument family used in U.K. renewables auctions and in the E.U.’s emerging SAF support schemes. Re-packaging only the risk component of the statutory credit as a floor would hold planner output fixed and save roughly \$0.5 B/yr in expected outlay; the producer-side counterpart of this claim, including the project-finance channel through which a floor also relaxes the participation constraint of a leveraged developer, is established in [Li and Yu \(2026\)](#).

5.7. Topological response to SAF price

The topological response of the supply chain to SAF wholesale price across the 20 KDE-discretized scenarios (computed on the *fixed-demand* model’s price sweep; Section 3.5, variant attribution) is shown in Figure 8 (in Section F). Panel (a) plots the present-value chain surplus $PVC(\omega)$ against $p_{SAF}(\omega)$, with KDE probability bars overlaid; the surplus turns positive between the \$5.53 and \$5.82 bins (interpolated crossing $\approx$ \$5.5/gal; the panel’s dashed marker is drawn at the first positive-surplus bin, \$5.82), below which $PVC < 0$ and the build-no-build decision tilts toward slack import. Panel (b) plots the realized facility counts $N_D(\omega)$ and $N_R(\omega)$ against p_{SAF} ; the topological expansion band over roughly \$5.8–\$6.6/gal is the range across which the chain adds marginal depot/refinery capacity beyond the deterministic baseline siting (depot count rising from ~ 225 to ≥ 270 across the band). The empirical implication is that policy-uncertainty in this price neighborhood is particularly influential on long-run capacity, a sharper version of the KDE-calibrated $\sigma_{KDE}^* = \$0.52/\text{gal}$ subsidy result in Section 2.4, because the marginal expansion there is at the steepest point of the surplus-vs-price curve.

The high-price expansion in panel (b) is a property of the fixed-demand chain’s revenue objective, and it does not contradict the near-zero high-price regret of Section 5.2: under the fixed-demand sweep, a higher exogenous price mechanically raises the marginal revenue of every additional depot, whereas under the endogenous-demand welfare objective the consumer-surplus integral flattens the marginal return to capacity above the central scale, and the certified wait-and-see sitings never exceed 248/34 in any bin (and equal it from bin 10 upward). The two panels should therefore be read as the revenue-side response; the welfare-side response is Figure 4.

Carbon in the MILP: what remains open. The decomposition of Section 5.5 prices the Pigouvian component analytically at the make-or-buy margin (Proposition 1); what it does not deliver is the *siting response* to a carbon price — the marginal-abatement-cost schedule from Pigouvian re-solves of the full MILP at $p_C \in \{50, 100, 200\}$ \$/tCO₂, together with 45Z producer-credit counterfactuals (status quo, removed, doubled). Those runs are left to follow-up work using the same model architecture.

5.8. Risk aversion (CVaR sensitivity)

A risk evaluation on the endogenous-demand variant places the committed central plan (248 depots, 34 refineries; $\mathbb{E}[L] = \$299$ M/yr, $CVaR_{0.95}[L] = \$921$ M/yr) against the wait-and-see topologies obtained from the refreshed low-price solves of Section C, vintage (ii): 192/26, 195/26, 197/27, 204/28, 211/30, and 248/34 at bins 1–6 respectively (solve gaps 1.3–4.8%; per-plan statistics and provenance in Table 9). The bin-6 solve returns the committed siting itself — identical depot and refinery sets — so the pool contains six distinct plans. The pool is deliberately broader than the certified wait-and-see frontier of Section 5.2: certification raised bins 2 and 4–9 to the 197/27 topology, so 195/26, 204/28, and 211/30 are no longer scenario-optimal in any bin; they are retained as first-stage candidates because enlarging the pool can only tighten the certification of the wait-and-see benchmarks, and can only weaken, never strengthen, the dominance claims below. Each plan is held fixed as a first-stage commitment and evaluated by fixed-siting recourse solves across all 20 KDE bins under a single data state (recorded gaps below 10^{-12}), with every wait-and-see benchmark certified against this pool (Section 3.5). Expectation and regret on one side, and the welfare-level tail on the other, rank these plans in opposite directions. Three findings establish that tension, and a grid-refinement lesson disciplines all three.

First, *provenance*: the committed plan’s own provenance is the deterministic bin-9 sweep, which terminated at an 18.1% gap on the fixed-demand present-value objective; its siting is, however, reproduced exactly by the bin-6 endogenous-demand wait-and-see solve at a 4.8% gap, so the committed topology is itself a converged candidate on the objective used here. We still do not solve the joint mean–CVaR MILP at any $\lambda > 0$, and no mean–CVaR optimum is claimed; the set contains no plan larger than 248/34 because no wait-and-see re-solve, warm-started from either the committed or a contracted topology, returned a larger siting in any bin.

Second, the *grid* matters, and we report the refined numbers. All cross-plan statistics here are recomputed at an 81-breakpoint SOS2 grid (LP-exact per bin), after a 41/81-breakpoint refinement exposed the 21-breakpoint evaluation’s cross-plan margins as chord-error artifacts: the recorded +\$15 M/yr expected-welfare advantage of the 197/27 plan reverses to –\$99.5 M/yr on refinement, with the 41- and 81-breakpoint values agreeing to \$0.1 M (the $O(\Delta Q^2)$ convergence signature). The committed plan’s own statistics barely move ($\mathbb{E}[L]$: \$299 to \$300 M/yr; $\text{CVaR}_{0.95}[L]$: \$921 to \$931 M/yr; tail-to-mean 3.1), so Result 1 is grid-robust; the cross-plan comparison at the coarse grid was not, exactly the failure mode the linearization invites on margins of this size.

Third, on the refined grid the committed plan minimizes *both* regret dimensions and maximizes expected welfare ($\mathbb{E}[L] = \$300$ against the 197/27 plan’s \$400 M/yr; $\text{CVaR}_{0.95}[L] = \931 against \$1,837 M/yr; $\mathbb{E}[W] = \$17.004$ against \$16.904 B/yr), while the 197/27 plan retains the welfare-level tail: its $\text{CVaR}_{0.95}[W]$ is higher by \$0.93 B/yr, a margin that is itself grid-robust (\$0.92 B at the coarse grid). The mean–CVaR preference therefore switches at $\lambda^* \approx 0.10$: below it the committed plan is preferred, above it the contracted plan is — by \$0.41 B/yr at the midpoint $\lambda = 0.5$.

One arithmetic identity links the regret and expectation findings and should not be double-counted: because the certified wait-and-see column coincides with the pool maximum in every bin (Section 5.2), the \$99.5 M/yr expected-welfare margin *equals* the expected-regret difference \$400 – \$300 M/yr — the two rankings share one expected-value number and part ways only in the tails.

The two objects disagree by construction, not by accident. Regret nets each plan against a scenario-specific ideal, so the smaller plan is charged for the upside it forgoes at high prices even though its welfare there remains within a few percent of the committed plan’s. Welfare levels instead face the tail where it lives, the lowest-price bins, and there the smaller plan’s avoided capital is a pure gain. For the commitment question of Section 5.2 the regret object is the right lens. For the instrument question of Theorem 1, what a tail-averse planner builds and what a subsidy must restore, the welfare-level object governs, and it says a planner with risk weight above $\lambda^* \approx 0.10$ builds smaller than the deterministic central plan.

The committed plan is the deterministic solve at the near-mean calibration bin, an expected-value (EEV) plan in the sense of (5) up to the \$0.08/gal difference between the bin-9 conditional mean and the distribution mean. On the refined grid it is itself the best expected-welfare candidate, so the candidate-based estimate of the value of the stochastic solution on this accounting is zero; what tail aversion prices is the \$0.93 B/yr tail shortfall. (The fixed-demand PVC accounting of Section 5.3 reaches the same conclusion on the expectation margin; the two accountings now agree on the marginal plan in expectation.) That the tail-preferred plan is a one-fifth contraction of the committed one ($197/248 = 0.79$ on depots, $27/34 = 0.79$ on refineries) is the model-scale counterpart of the contraction that Theorem 1 prices in closed form. A non-degenerate welfare-level comparison at intermediate plans, and any claim about the true mean–CVaR optimum, still requires the joint MILP, which remains open.

What the closed form contributes meanwhile is structure. Theorem 1 predicts $\sigma^*(\lambda)$ *linear* in λ in the collapsed

regime $\alpha \geq \pi_H$, with slope $\pi_H(a_H - a_L) = 0.472 \times 2.194 = \$1.04/\text{gal}$ per unit λ under the KDE-anchored calibration. Whether the full 700k-variable MILP at intermediate λ would deliver the same linearity, or exhibit integer kinks at depot-count transitions that the two-state model abstracts away, is an open empirical question that we flag in Limitations as a candidate for follow-up investigation.

Cross-checking the closed form against the MILP. The closed form gives the committed-plan-versus-contraction comparison a theoretical anchor, and the MILP now disciplines the closed form’s two free cost parameters. At interior optima the wait-and-see solves trace the toy’s first-order condition $a_\omega = c + K Q_\omega$, so regressing the per-bin demand intercept on realized wait-and-see output identifies both jointly: across all 20 bins, $K = 4.3 \times 10^{-9} \$/\text{gal}^2$ and $c = \$5.00/\text{gal}$ ($R^2 = 0.68$), with implied capacity curvature $k = K - b = 3.1 \times 10^{-10}$, small relative to the demand slope. Subsample fits are unstable — bins 1–9 give $K = 2.0 \times 10^{-9}$, and bins 10–20, where output is recourse-constrained by the fixed 248/34 topology, admit no sensible fit — which is direct evidence of the integer kinks that make the MILP’s effective curvature piecewise rather than constant; we carry that instability as the calibration’s uncertainty band.

Three checks follow. First, the *contraction*: in the collapsed regime the $\lambda = 1$ optimum has effective intercept A_L (Section A, equation (34)), so the tail-only solution is the deterministic optimum at the low state. With the calibrated c , the predicted contraction in the endogenous-demand intercept space is $(a_{\text{bin1}} - c)/(\bar{a} - c) = 0.84$, against the realized bin-1 wait-and-see contraction of $192/248 = 0.774$ on depots and $26/34 = 0.765$ on refineries: the MILP contracts somewhat more than the smooth prediction, consistent with minimum-throughput lumpiness shedding whole facilities at the margin. Second, the *information-value reconciliation* that an earlier draft declined is now possible: $\text{EVPI} = \text{Var}(A)/(2K) = \190 M/yr at the calibrated curvature, the same order as the $\$255 \text{ M/yr}$ candidate-based chain-surplus estimate and the $\$299 \text{ M/yr}$ best-in-pool regret. The three accountings differ in construction and are not expected to coincide, but they agree in magnitude, which is what Corollary 2 can deliver on a piecewise-curvature problem. Third, the *benefit–cost content* of Corollary 4 reported in Section 5.6: the recovered deadweight loss $\sigma^{*2}/(2K) = \$31\text{--}\65 M/yr sits an order of magnitude below the flat credit’s MCPF cost.

A two-state injection test of the credit. A direct test of Theorem 1 on the integer chain is to inject the credit into the planner’s program and ask whether it undoes the risk distortion. The full 20-bin mean–CVaR extensive form is beyond reach at this instance size, so we run the test on the model’s own two-state collapse (the exact structure Theorem 1 prices; state intercepts and probabilities from Section 2.4), in two layers. First, free extensive-form solves at $\lambda = 0$, at $\lambda = 0.5$, and at $\lambda = 0.5$ with a $\$0.518/\text{gal}$ credit all return the committed 248/34 topology at realized gaps of 3.0–4.1%: the risk perturbation’s predicted output effect ($\sigma^*/K \approx 0.12 \text{ B gal}$, 3.7% of output; a welfare effect on the order of 0.1–0.2%) sits below the resolvable tolerance of the free MILP, which is the precise sense in which the joint mean–CVaR program remains open (Section 6.4). Second, fixing the sitings makes the test exact: LP evaluations of the committed plan and the two leading contracted candidates under the same three configurations, at the refined 81-breakpoint grid, resolve the ranking to machine precision. At $\lambda = 0$ the two-state ordering is $197/27 \succ 248/34 \succ 192/26$, with the committed plan $\$30.8 \text{ M/yr}$ behind the leader. At $\lambda = 0.5$ without the credit, risk aversion demotes the committed plan to last and widens its wedge to $\$306.9 \text{ M/yr}$. Adding σ^* restores the risk-neutral ordering exactly and closes the wedge to $\$47.8 \text{ M/yr}$: the credit undoes 94% of the risk-induced distortion in the plan comparison (97% at the coarse grid — the restoration share is grid-robust).

One caveat keeps the two-state test in its lane: collapsing twenty bins to two compresses the high-price upside that favors the large plan, so plan *levels* are not transportable from the collapse to the full bin set (where the committed plan leads expected welfare; Table 9) — the test speaks to the λ - and credit-comparative statics, which is what Theorem 1 asserts, not to the level ranking.

A boundary caveat carries over: the fixed-demand model’s interpolated build-vs-slack crossing sits at $\approx \$5.5/\text{gal}$, so the low-price states are near the no-build margin, whereas the toy’s Assumption 5 keeps low-state production interior; the toy’s per-gallon policy objects (σ^* , the floor strike, the outlay ratios) are first-order summaries away from that margin, not at it. The mean–CVaR objective that generates σ^* is validated on the MILP through the calibrated contraction and the regret statistics of Section 5.2; a full MILP re-solve reporting CVaR of welfare levels is left to the joint mean–CVaR work flagged above.

6. Discussion and Robustness

This section examines the partial-equilibrium scope of the welfare functional, sensitivity to ash quality and county-level aggregation, and the modeling restrictions that bound the σ^* benchmark’s interpretation.

6.1. Partial-equilibrium scope and demand-elasticity choices

The welfare functional in (3) is a *partial-equilibrium* (PE) construct that excludes four general-equilibrium (GE) channels the broader biofuel welfare literature has identified as quantitatively material: (i) computable general-equilibrium (CGE) stumpage-price endogeneity in response to bioenergy demand (Hertel et al. 2010): we hold stumpage exogenous and treat residue as a by-product of mill demand (USDA Forest Service 2023a), understating the upward response to a 3.0 B gal/yr SAF mandate; (ii) carbon-intensity-based market segmentation and LCFS rents: Holland, Hughes, and Knittel (2009) show an intensity standard is never first-best and can even raise emissions, with implied average abatement costs of \$307–\$2,272 per tCO₂e in their central cases (against a cost-effective alternative range of \$60–\$868), and Khanna, Rajagopal, and Zilberman (2021) document analogous pass-through and welfare wedges across the biofuel policy stack; (iii) partial RIN pass-through under the Renewable Fuel Standard (Lade and Bushnell 2019); and (iv) indirect land-use change (Searchinger et al. 2008). The channels do not all push in the same direction, and the decomposition of Section 5.5 clarifies which component each one moves.

Channel (i) operates through two routes. On the cost route, an endogenous stumpage response raises the chain’s marginal cost at scale, which moves the level objects (equilibrium output, welfare, private participation) but *not* the risk component, whose closed form $\lambda\pi_H(a_H - a_L)$ contains no cost term. To the extent stumpage prices *co-move* with fuel prices, the effect on σ^* is in fact downward: the closed form prices the spread of the net premium $A(\omega) = a(\omega) - c(\omega)$, and positive covariance between cost and price compresses that spread (a natural hedge). On the emissions route, stumpage-driven land-use carbon debt erodes Δe ; the oft-cited 25–50 gCO₂e/MJ magnitude is a dedicated-plantation number, and for harvest residues, a by-product of sawtimber-driven rotations, the marginal debt is an order of magnitude smaller, entering only through the stumpage-price feedback. We state the exposure plainly: were the plantation-scale upper end to apply, Δe would fall by 3.3–6.6 kgCO₂e/gal and the between-the-vintages reading of Section 5.5 would fail; the residue-specific reading keeps the erosion inside the $\pm 20\%$ band reported there.

Channels (ii)–(iii) operate on the carbon component’s *incidence*: LCFS credit values ($\sim \$0.20$ – $\$0.35/\text{gal}$ at recent settlement prices) and RIN value passed through at 50–70% are existing, partial market payments for the same avoided carbon. Because our price proxy is fitted on fossil jet-A, none of that stacked value is in $a(\omega)$, and the risk component σ^* is unaffected either way (Proposition 1(iii)). Channel (iv)’s 0–110 g CO₂e/MJ corn-ethanol indirect-land-use-change (iLUC) bracket is largely shut off for residue feedstock because residues are by-products of timber harvest rather than dedicated bioenergy plantings; iLUC contributes only through channel (i)’s stumpage feedback.

The net implication for the decomposition is asymmetric across components. The risk component $\sigma_{\text{KDE}}^* = \$0.52/\text{gal}$ is a lower bound in the spread-conditional sense of Section 5.4 and an upper bound in offtake coverage (the offtake-coverage paragraph below). On the carbon side, stacking works in the direction of a *higher* implied SCC: where a producer collects LCFS/RIN value on top of 45Z, total policy support per gallon exceeds the federal credit alone, and the SCC required to rationalize the full stack rises: at $\$0.30/\text{gal}$ of stacked value, from $\$77$ – $\$91$ to approximately $\$106$ – $\$121/\text{tCO}_2$. The $\$77$ – $\$91$ band of Section 5.5 should therefore be read as the federal-credit-only figure. A CGE-consistent welfare evaluation is the natural follow-up direction flagged in the Limitations subsection.

Demand elasticity. Both the toy benchmark of Section 2.4 and the endogenous-demand variant of Section 3.5 use linear demand $P^d(Q; \omega) = a(\omega) - bQ$ with $b > 0$; the fixed-demand model of Section 3 is the $b \rightarrow 0^+$ limit, retained as a computational simplification for the deterministic siting and the information-value accounting, not as an empirical claim about SAF demand. Two facts discipline what the elasticity choice can and cannot affect. First, the headline risk objects are elasticity-invariant by construction: σ^* and the instrument ranking depend only on the KDE primitives and λ (Section 4.4), and the first-stage topology is unchanged across the grid, so the elasticity choice moves only the level of the regret statistics. Second, no direct estimate of a long-run SAF demand elasticity exists, so the grid $\{-0.3, -0.5, -1.0\}$ brackets rather than estimates. The elastic edge is loosely anchored on the closest observed fuel-substitution behavior — household flex-fuel switching in Brazil (Salvo and Huse 2013) — whose retail spot mechanism and monthly horizon make it at best an upper bound for a decade-horizon capacity decision; that is why the grid stops at -1.0 rather than adopting the -2 to -3 switching estimates. The inelastic edge reflects the long-tenor (5–15-year) offtake contracts that dominate the U.S. SAF buyer side. Short-run jet-fuel demand under contract is far more inelastic still ($\varepsilon \sim -0.15$ in EIA Short-Term Energy Outlook (STEO) regressions, similar to airline fuel demand in Hofer, Dresner, and Windle 2010); that estimate answers a different question — the within-contract quantity response to a price shock — whereas a ten-year siting decision responds to the long-run schedule at which buyers are willing to contract, and the grid is a bracket on that schedule. The operative calibration for the regret evaluation is the grid midpoint -0.5 (Section 3.5).

Offtake coverage cuts the other way. The same long-tenor contracts that justify the inelastic side of the grid also bound the risk component from *above*. The spread $a_H - a_L = \$2.19/\text{gal}$ is inherited from the constructed revenue-price dispersion — joint fuel-price and RIN-price variation (Section 4); if a share θ of output is sold under fixed-price offtake, the committing agent’s effective spread is $(1 - \theta)(a_H - a_L)$ and the risk component scales linearly: $\sigma^*(\theta) = (1 - \theta) \sigma^*$. At $\theta = 0.5$, $\sigma^* = \$0.26/\text{gal}$ and the implied-SCC band of Section 5.5 rises to $\$102$ – $\$117/\text{tCO}_2$, still between the federal vintages. We report the $\theta = 0$ case because contract coverage in a

scaled 3 B gal/yr market is unobservable today, but the directional statement matters for honesty: σ^* is a *lower* bound in the financing-friction and hurdle-wedge channels (which act on λ) and an *upper* bound in three others: offtake coverage, cost–price co-movement (the natural hedge above), and partial financial hedging (Section 5.6). The conservativeness claims in this paper should be read with both signs in view. One boundary is worth stating: even at full coverage ($\theta = 1$, $\sigma^* = 0$) the implied-SCC band is $(\$1.30, \$1.45)/\Delta e = \$127\text{--}\$142/\text{tCO}_2$, still between the federal vintages, so the between-the-vintages reading of Section 5.5 survives *any* degree of offtake coverage.

Outside-option price. The slack-import price enters the welfare objective only through unmet demand, so any solution that meets the demand floor from domestic production has a p^{jet} -invariant objective; the deterministic baseline builds with zero slack, and its chain surplus is unchanged at any p^{jet} . What p^{jet} moves is the build-versus-slack margin: producing a marginal gallon earns p_{SAF} and avoids importing at p^{jet} , so the margin shifts one-for-one with the outside option — lowering p^{jet} from the \$3.50 benchmark to the 2025 range (\$2.10–\$2.60) raises the revenue price at which marginal capacity breaks even by \$0.90–\$1.40/gal. Warm-started re-solves of the crossing-region bins (6–11) at $p^{\text{jet}} \in \{2.10, 2.60\}$ confirm the two bounded statements this implies: the full-build incumbent dominates the all-import plan at every tested price (by \$4.4 B to \$9.8 B/yr of PVC), and its reported surplus is numerically identical across p^{jet} (zero slack in every incumbent). Whether intermediate topologies — a smaller chain plus imports — dominate full build at the lowest-price bins is not resolved at the realized gaps (14–141%; the middle ground between two warm-started extremes is unexplored) and moves with the same margin, so we report the build-vs-slack crossing as p^{jet} -dependent in the one-for-one analytic sense rather than claim a solved sweep.

6.2. Sensitivity to ash quality

The 10-tier ash-yield discretization yields a $\pm\$0.30/\text{gal}$ welfare-equivalent SAF-cost range at fixed $p_{\text{SAF}} = \$5.97/\text{gal}$, with the highest-ash tier reducing yield by $\sim 12\%$ relative to the median and the lowest-ash tier raising yield by $\sim 9\%$. Ash quality operates on the cost/yield side, so by the cost-invariance of the closed form it does not move σ^* at all; its materiality is for the welfare and cost *levels*, where the ash-induced per-gallon cost range $((\Delta\phi/\phi) \cdot p_{\text{SAF}} \approx \$0.18/\text{gal}$, within the $\pm\$0.30/\text{gal}$ band above) is an order of magnitude smaller than the demand-state spread $a_H - a_L \approx \$2.19/\text{gal}$ that drives the risk results.

6.3. Aggregation level

The Stage-2 operational solve runs at the FIPS-5 county aggregation level (five-digit Federal Information Processing Standards county codes). Internal validation against the underlying 50 mi² hex resolution returns +0.15% aggregate-NPV agreement, –0.07% total-residue agreement, and 0.00% agreement on total stumpage cost (TSC) (Table 7); per-state re-allocation can shift by up to $\pm 10\%$ but the total and the state Top-3 ranking are preserved. Construction details are in Section B.

6.4. Limitations

Seven modeling simplifications constrain the present scope.

- (i) *Single-period investment timing.* Depot and refinery decisions are made once at $t = 0$ and held fixed across the 10-year horizon. Holding all 34 refineries at $t = 0$ overstates first-period CAPEX exposure relative to a 2027–2030 / 2030–2035 / 2035–2040 staged buildout; the real-options deferral value is bounded above by the value of perfect information (Boomsma, Meade, and Fleten 2012), an object our candidate-based estimate approximates but does not certify (Section D), so we do not attach a numeric bound. A multi-period real-options extension—where the planner can defer infrastructure commitment in response to evolving SAF-price signals—is a natural follow-up.
- (ii) *Exogenous mill demand.* Mill demand for hardwood pulp/sawtimber is taken exogenously; in reality, residue supply curves and pulpwood demand co-vary, and a Faustmann-style supply-side response is left to future work.
- (iii) *County-level aggregation.* County-level aggregation washes out within-county heterogeneity; Proposition 3 preserves the harvest-cost term under its stated hypotheses, and the full accounting stack matches across resolutions to within $\pm 0.15\%$ (Table 7).
- (iv) *Externality outside the MILP.* The stochastic program is solved at $p_C = 0$; the Pigouvian component of the credit is priced analytically at the make-or-buy margin via Proposition 1 (Section 5.5), which is exact in the closed form and first-order in the MILP. What the analytical route omits is the siting response to a carbon price—the endogenous marginal-abatement-cost schedule—which requires Pigouvian re-solves left to future work.
- (v) *Asymmetric substitution structure.* Perfect SAF/fossil-jet substitution (within ASTM blend limits) is assumed only on the *slack-import* side: unmet demand is covered by fossil jet at p^{jet} , which gives shortfall an opportunity cost. The *revenue* side treats SAF as a distinct, contract-driven market whose price distribution is the constructed drop-in fuel-plus-credit series of Section 4.3, on a different basis from p^{jet} (Section C); i.e., the two fuels are substitutes for the buyer of last resort but not for the ESG-mandated offtaker. Unifying the two sides in a constant-elasticity-of-substitution (CES) nest would introduce a substitution-elasticity parameter that the present calibration cannot identify.
- (vi) *Risk-averse first stage not optimized.* The mean–CVaR program (4) is evaluated over a fixed six-plan candidate set of wait-and-see topologies spanning two evaluation vintages (Section 5.8), not solved as a joint MILP. The two-state injection test of Section 5.8 shows why: the λ perturbation’s welfare effect sits below the free MILP’s attainable tolerance, so the joint program is evaluated exactly only on fixed candidate sitings. Whether the joint mean–CVaR optimum at intermediate λ tracks the linear $\sigma^*(\lambda)$ schedule of Theorem 1 or exhibits integer kinks at depot-count transitions remains open at full 20-bin scale, as are converged low-price wait-and-see solves at the $\varepsilon = -1.0$ calibration (Section 3.5).
- (vii) *Price-state persistence.* The scenario structure treats the realized price as a persistent decade-scale state. Under decade-scale mean reversion the committed chain earns a multi-year average price whose dispersion is much smaller, and the risk component falls from \$0.52 toward \$0.11/gal (block-bootstrap evidence in Section C); the sample is too short to pin the long-run persistence of the fuel-plus-RIN price level, so we carry the persistence assumption as the primary calibration uncertainty on the risk side.

Together these simplifications bound the interpretation rather than sign it. The multi-period extension would

raise the EVPI (deferral value is bounded above by it); the elasticity direction is undetermined, since the $\varepsilon = -1.0$ low-price solves do not converge (Section 3.5); and on σ^* the channels run in both directions and are enumerated with both signs in Section 6.1.

7. Conclusion

We have decomposed the welfare case for the Section 45Z SAF credit into two statute-relevant pieces, the carbon externality and price risk under irreversible investment, in a two-stage stochastic welfare-maximization model of the SE-U.S. forest-residue supply chain, and asked whether a flat per-gallon credit is the right instrument for the risk component. The closed-form benchmark Theorem 1 maps the empirical price distribution into a risk-premium subsidy of $\sigma^* = \$0.52/\text{gal}$ at the baseline risk-aversion weight; the underlying tail-loss and information-value primitives are $\text{CVaR}_{0.95}[L] = \921 M/yr on the certified evaluation and a candidate-based EVPI estimate of $\$255 \text{ M/yr}$ on the fixed-demand chain-surplus accounting.

The results speak to policy on two fronts: the credit’s level and its form. On the *level*, the decomposition is exactly additive (Proposition 1), and carbon plus risk together rationalize the statutory $\$1.30\text{--}\$1.45/\text{gal}$ credit at an implied social cost of carbon of $\$77\text{--}\$91/\text{tCO}_2$ under the baseline reading — as-enacted rate, flat-credit delivery, incremental to existing RFS support — between the two federal SCC vintages. The implied SCC is, however, a schedule rather than a number (Table 6): reasonable choices of instrument form, statutory vintage, emissions basis, and RIN capitalization move it from $\$22$ to roughly $\$382/\text{tCO}_2$, and the post-2025 rate actually in force prices carbon below the IWG floor. The familiar question “is 45Z too generous?” has no vintage-independent answer: at the EPA (2023) SCC the carbon channel alone ($\$1.94/\text{gal}$) exceeds the statutory credit, while at the IWG (2021) interim value the credit requires substantial risk aversion ($\lambda \approx 0.75\text{--}0.90$) to justify. The rationalization frontier of Figure 5 makes the trade-off explicit rather than leaving the residual unattributed. On the *form*, the risk component is a payment for variance, and a flat credit purchases it at par in every state. A one-sided price floor delivers the identical capacity correction at 69% of the flat credit’s expected fiscal outlay, and a two-sided contract-for-difference delivers it at zero expected outlay (Proposition 2); with costly public funds the outlay gap is a strict welfare wedge (Corollary 4). The design implication is a mixed instrument: pay the Pigouvian component flat, and pay the risk component through the floor/CfD family already standard in European renewables procurement. Both implications rest on assumptions worth stating plainly.

The interpretation is qualified along two dimensions: scope and data. On scope, the analysis is partial equilibrium. The risk component’s closed form is invariant to cost-level feedbacks; omitted channels raise it to the extent they widen the price spread the committing agent faces or its effective risk aversion (financing frictions, hurdle wedges), in which case σ^* is a lower bound and the implied-SCC band shifts down as those channels are added; offtake coverage, cost–price co-movement, and partial hedging work in the opposite direction, scaling σ^* down and the implied-SCC band up, though even at full coverage the band stays between the federal vintages (Section 6.1). On data, the SAF spot market is thin and we proxy the relevant price by wholesale jet-fuel data plus an SAF premium, which makes the EVPI level conservative relative to a deeper SAF-price calibration when one becomes available.

This framework points toward four extensions. First, solving the joint mean–CVaR MILP at intermediate λ , rather than evaluating (4) over a candidate set, would test whether the integer chain tracks the linear $\sigma^*(\lambda)$ schedule of Theorem 1 or exhibits kinks at depot-count transitions. Second, a multi-period real-options specification

(Dixit and Pindyck 1994; Boomsma, Meade, and Fleten 2012; Insley 2017) would replace the one-shot capacity commitment with staged buildout and capture the deferral value the present framework bounds above by the EVPI. Third, embedding an endogenous stumpage response (Faustmann 1849; Amacher, Ollikainen, and Koskela 2009) would relax the partial-equilibrium scope and bring general-equilibrium feedbacks into the welfare object. Fourth, the Pigouvian component is priced here at the make-or-buy margin; Pigouvian re-solves of the MILP at $p_C \in \{50, 100, 200\}$ \$/tCO₂ would deliver the endogenous-siting marginal-abatement-cost schedule, and explicit 45Z incidence (Lohawala, Toman, and Joiner 2024) together with the producer-side project-finance channel of Li and Yu (2026) would extend the planner-side decomposition to the private participation margin.

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A. Proofs of the closed-form benchmark

This appendix collects the proofs of Lemma 1, Theorem 1, Corollaries 2 to 4, and Propositions 1 and 2.

Throughout, define the net demand premium $A \equiv a(\omega) - c$ with realizations $A_H \equiv a_H - c$ and $A_L \equiv a_L - c$. By Assumption 5, $A_L > 0$; by Assumption 4, $A_H > A_L$. The combined curvature constant is $K \equiv b + k > 0$ under Assumptions 2 and 3.

A.1. Lemma 1 (strict concavity + interior optimum)

PROOF OF LEMMA 1. *Step 1: Strict concavity.* The per-state social welfare under Assumptions 2 and 3 is

$$(28) \quad W(Q; \omega) = (a(\omega) - c)Q - \frac{K}{2}Q^2, \quad K \equiv b + k.$$

Direct differentiation gives

$$(29) \quad \frac{\partial^2 W(Q; \omega)}{\partial Q^2} = -K < 0,$$

since $b > 0$ and $k > 0$. The planner’s mean–CVaR functional is

$$(30) \quad \mathcal{W}(Q) = (1 - \lambda) \mathbb{E}[W(Q; \omega)] + \lambda \text{CVaR}_\alpha[W(Q; \omega)].$$

The expectation $\mathbb{E}[W(Q; \omega)] = \pi_H W(Q; H) + \pi_L W(Q; L)$ is a convex combination of the per-state welfares, each strictly concave by (29), hence strictly concave in Q . Under the discrete two-state representation, $\text{CVaR}_\alpha[W(Q; \omega)]$ is itself a convex combination of the per-state welfares (with non-negative weights summing to one — explicit formulae appear in (32) and (33) below), hence also strictly concave in Q . A convex combination of strictly concave functions is strictly concave, so $\mathcal{W}(Q)$ is strictly concave on $\mathbb{R}_+$ with second derivative $-K < 0$.

Step 2: Interior optimum. Writing $\mathcal{W}(Q) = A_{\text{eff}}Q - \frac{K}{2}Q^2$ where A_{eff} is the effective demand intercept

derived in Step 3 of the proof of Theorem 1 (equations (34) and (35) below), the first-order condition gives

$$(31) \quad Q_{\text{risk}}^* = \frac{A_{\text{eff}}}{K}.$$

We show $A_{\text{eff}} > 0$ for all $\lambda \in [0, 1]$ and $\alpha \in (0, 1)$.

At $\lambda = 0$ both regime expressions reduce to $A_{\text{eff}} = \mathbb{E}[A] = \pi_H A_H + \pi_L A_L > A_L > 0$ by Assumptions 1, 4 and 5.

At $\lambda = 1$ in the collapsed regime ($\alpha \geq \pi_H$), (34) gives $A_{\text{eff}}|_{\lambda=1} = A_L > 0$. In the split regime ($\alpha < \pi_H$), (35) gives $A_{\text{eff}}|_{\lambda=1} = \mathbb{E}[A] - \frac{\alpha \pi_L}{1-\alpha} (a_H - a_L)$; expanding $\mathbb{E}[A] = \pi_H A_H + \pi_L A_L$ and using $\pi_H (a_H - a_L) = \mathbb{E}[A] - A_L$ yields $A_{\text{eff}}|_{\lambda=1} = A_L + \frac{(\pi_H - \alpha)}{1-\alpha} (a_H - a_L) > A_L > 0$ since $\alpha < \pi_H$.

For intermediate $\lambda \in (0, 1)$, A_{eff} is linear in λ with $\partial A_{\text{eff}} / \partial \lambda = -\pi_H (a_H - a_L) < 0$ in the collapsed regime and $-\frac{\alpha \pi_L}{1-\alpha} (a_H - a_L) < 0$ in the split regime; combined with the strictly positive endpoint values above, $A_{\text{eff}}(\lambda) \geq \min\{A_{\text{eff}}|_{\lambda=0}, A_{\text{eff}}|_{\lambda=1}\} > 0$.

Hence $Q_{\text{risk}}^* > 0$ is interior, and the non-negativity constraint $Q \geq 0$ is slack at the optimum — no Lagrangian–KKT bookkeeping is required.

Step 3: Sufficiency of the FOC. Strict concavity of $\mathcal{W}$ (Step 1) combined with the existence of an interior critical point (Step 2) implies the FOC is both necessary and sufficient for a global maximum, and that maximum is unique. $\square$

A.2. Theorem 1 (piecewise σ^* closed form)

PROOF OF THEOREM 1. *Step 1: Per-state welfare.* By the area-under-demand-minus-cost integral with linear inverse demand $p(Q; \omega) = a(\omega) - bQ$ (Assumption 2) and quadratic investment cost $C(Q) = \frac{k}{2} Q^2$ (Assumption 3), the per-state social welfare is given by (28), reproduced here for reference:

$$W(Q; \omega) = (a(\omega) - c) Q - \frac{K}{2} Q^2.$$

Step 2: CVaR representation — two regimes. By the Rockafellar–Uryasev auxiliary-function representation (Rockafellar and Uryasev 2000), written in welfare form (equivalent to the loss-CVaR form $\inf_{\zeta} \{\zeta + (1 - \alpha)^{-1} \mathbb{E}[(L - \zeta)_+]\}$ under the substitution $L = -W$),

$$\text{CVaR}_\alpha[W(Q; \omega)] = \sup_{\zeta \in \mathbb{R}} \left\{ \zeta - \frac{1}{1-\alpha} \mathbb{E}[(\zeta - W(Q; \omega))_+] \right\},$$

which is the conditional mean of W in its worst- $1 - \alpha$ tail. The denominator $1 - \alpha$ is finite by Assumption 6. By Assumption 4 ($a_H > a_L$), the bad-tail event is unambiguously state L , since $W(Q; L) < W(Q; H)$ at any $Q > 0$. Two regimes arise from where the tail mass $1 - \alpha$ falls.

Regime A (collapsed): $\alpha \geq \pi_H$. The tail probability $1 - \alpha \leq \pi_L$ fits entirely inside state L , so the discrete CVaR collapses to the state- L welfare:

$$(32) \quad \text{CVaR}_\alpha[W(Q; \omega)] = W(Q; L) \quad (\alpha \geq \pi_H).$$

Regime B (split): $\alpha < \pi_H$. The tail mass exhausts all of state L (probability π_L) plus a slice of size

$1 - \alpha - \pi_L = \pi_H - \alpha$ of state H . The probability-weighted tail average is

$$(33) \quad \text{CVaR}_\alpha[W(Q; \omega)] = \frac{\pi_L W(Q; L) + (\pi_H - \alpha) W(Q; H)}{1 - \alpha} \quad (\alpha < \pi_H).$$

Step 3: Effective demand intercept. Substitute (32) or (33) into the mean-CVaR objective (30) and collect the Q -coefficient.

Regime A. The effective probability on state L is $\pi_L + \lambda\pi_H$ and on state H is $(1 - \lambda)\pi_H$. Hence

$$(34) \quad \begin{aligned} A_{\text{eff}}^{\text{coll}} &= (\pi_L + \lambda\pi_H)(a_L - c) + (1 - \lambda)\pi_H(a_H - c) \\ &= \pi_H(a_H - c) + \pi_L(a_L - c) - \lambda\pi_H[(a_H - c) - (a_L - c)] \\ &= \mathbb{E}[A] - \lambda\pi_H(a_H - a_L). \end{aligned}$$

Regime B. The effective probability on state H becomes $(1 - \lambda)\pi_H + \lambda(\pi_H - \alpha)/(1 - \alpha)$, which simplifies to $\pi_H - \frac{\lambda\alpha\pi_L}{1 - \alpha}$ using $\pi_L = 1 - \pi_H$. The effective probability on state L is the complement, $\pi_L + \frac{\lambda\alpha\pi_L}{1 - \alpha}$. Collecting the Q -coefficient gives

$$(35) \quad A_{\text{eff}}^{\text{split}} = \mathbb{E}[A] - \lambda \frac{\alpha\pi_L}{1 - \alpha} (a_H - a_L).$$

Step 4: FOC and risk-averse optimum. By Lemma 1, the planner's objective $\mathcal{W}(Q) = A_{\text{eff}}Q - \frac{K}{2}Q^2$ is strictly concave in Q with interior maximum. The first-order condition $\partial\mathcal{W}/\partial Q = A_{\text{eff}} - KQ = 0$ delivers

$$(36) \quad Q_{\text{risk}}^* = \frac{A_{\text{eff}}}{K},$$

with A_{eff} given by (34) or (35) according to regime. The risk-neutral benchmark $\lambda = 0$ reduces both expressions to $A_{\text{eff}} = \mathbb{E}[A]$, so

$$(37) \quad Q_{\text{RN}}^* = \frac{\mathbb{E}[A]}{K}.$$

Step 5: Subsidy that restores the risk-neutral optimum. A per-unit production subsidy σ shifts the committing agent's effective intercept by σ : by Assumption 9, the credit enters the agent's marginal return identically to a price increment while its funding stays outside the mean-CVaR aggregate (were it financed inside the aggregate, the transfer would net out state by state and leave the optimum unchanged). The subsidy that restores Q_{RN}^* solves $A_{\text{eff}} + \sigma^* = \mathbb{E}[A]$. Substituting (34) and (35) yields

$$(38) \quad \sigma_{\text{coll}}^* = \lambda\pi_H(a_H - a_L) \quad (\alpha \geq \pi_H),$$

$$(39) \quad \sigma_{\text{split}}^*(\alpha) = \lambda \frac{\alpha\pi_L}{1 - \alpha} (a_H - a_L) \quad (\alpha < \pi_H).$$

Both are non-negative under Assumptions 4 and 7. Continuity at the boundary $\alpha = \pi_H$ is established in Corollary 3 below. Combining the two regimes into a single closed form,

$$(40) \quad \sigma^*(\alpha) = \lambda(a_H - a_L) \cdot \min\left\{\frac{\alpha\pi_L}{1 - \alpha}, \pi_H\right\}.$$

The risk-neutral limit $\lambda = 0$ gives $\sigma^* = 0$ for all α ; the fully tail-focused limit $\lambda = 1$ gives the upper bound $\sigma^*(\alpha) = \pi_H(a_H - a_L)$ in the collapsed regime. $\square$

A.3. Corollary 3 (continuity at $\alpha = \pi_H$)

PROOF OF COROLLARY 3. Evaluate the split-regime expression (39) at the boundary $\alpha = \pi_H$, using $1 - \pi_H = \pi_L$ from Assumption 1:

$$\sigma_{\text{split}}^*(\pi_H) = \lambda \frac{\pi_H \pi_L}{1 - \pi_H} (a_H - a_L) = \lambda \frac{\pi_H \pi_L}{\pi_L} (a_H - a_L) = \lambda \pi_H (a_H - a_L) = \sigma_{\text{coll}}^*.$$

Hence the left and right limits of $\sigma^*(\alpha)$ at $\alpha = \pi_H$ coincide, establishing continuity on $(0, 1)$.

The one-sided derivatives differ. Differentiating (39) with respect to α ,

$$\frac{d\sigma_{\text{split}}^*}{d\alpha} = \frac{\lambda \pi_L (a_H - a_L)}{(1 - \alpha)^2},$$

so the left derivative at the boundary is

$$(41) \quad \sigma^{*'}(\pi_H^-) = \frac{\lambda \pi_L (a_H - a_L)}{(1 - \pi_H)^2} = \frac{\lambda (a_H - a_L)}{\pi_L} > 0,$$

while (38) is constant in α , giving right derivative $\sigma^{*'}(\pi_H^+) = 0$. The mismatch (41) vs. 0 is a benign kink in $\sigma^*(\alpha)$, not a discontinuity. $\square$

A.4. Corollary 2 (EVPI = $\text{Var}(A)/(2K)$)

PROOF OF COROLLARY 2. We compute the expected value of perfect information for the risk-neutral planner ($\lambda = 0$), the textbook EVPI baseline used in Section 2.3 of the main text.

Step 1: Wait-and-see optimum per scenario. For each realized state ω , the per-state planner's problem $\max_Q W(Q; \omega)$ has FOC $(a(\omega) - c) - KQ = 0$, so the wait-and-see output is

$$(42) \quad Q^*(\omega) = \frac{a(\omega) - c}{K} = \frac{A}{K}.$$

By the envelope theorem (Mas-Colell, Whinston, and Green 1995, Theorem M.L.1), substituting (42) into (28) delivers the optimal value

$$(43) \quad W^*(\omega) = \frac{(a(\omega) - c)^2}{2K} = \frac{A^2}{2K}.$$

Step 2: Recourse optimum at the mean. The risk-neutral recourse plan uses $Q_{\text{RN}}^* = \mathbb{E}[A]/K$ from (37). Substituting into $\mathbb{E}_\omega[W(Q_{\text{RN}}^*; \omega)] = \mathbb{E}[A] \cdot Q_{\text{RN}}^* - \frac{K}{2}(Q_{\text{RN}}^*)^2$ yields

$$(44) \quad W^*(\mathbb{E}[A]) = \frac{(\mathbb{E}[A])^2}{2K}.$$

(Use linearity of expectation: $\mathbb{E}[A \cdot Q_{\text{RN}}^*] = Q_{\text{RN}}^* \mathbb{E}[A] = (\mathbb{E}[A])^2/K$.)

Step 3: $EVPI = \text{Var}(A)/(2K)$. Following the Madansky (1960) inequality framework, the expected value of perfect information is

$$EVPI \equiv \mathbb{E}_\omega[W^*(\omega)] - W^*(\mathbb{E}[A]).$$

Substituting (43) and (44),

$$\begin{aligned} EVPI &= \frac{\mathbb{E}[A^2]}{2K} - \frac{(\mathbb{E}[A])^2}{2K} \\ &= \frac{\mathbb{E}[A^2] - (\mathbb{E}[A])^2}{2K} \\ &= \frac{\text{Var}(A)}{2K}. \end{aligned} \tag{45}$$

The display in (45) matches the textbook quadratic-objective expression in Birge and Louveaux (2011, §4.7).

Specialized to the two-state distribution under Assumption 1, $\text{Var}(A) = \pi_H \pi_L (a_H - a_L)^2$, so

$$EVPI = \frac{\pi_H \pi_L (a_H - a_L)^2}{2K}. \tag{46}$$

□

A.5. Proposition 1 (additive credit decomposition)

Throughout this subsection and the next, write $m_\alpha \equiv \min\{\alpha\pi_L/(1-\alpha), \pi_H\}$, so that the two regime expressions (34) and (35) combine into the single form

$$A_{\text{eff}} = \mathbb{E}[A] - \lambda m_\alpha (a_H - a_L), \tag{47}$$

and $\sigma^*(\alpha) = \lambda m_\alpha (a_H - a_L)$ by (40).

PROOF OF PROPOSITION 1. *Step 1: First-best output.* With the externality internalized, per-state social welfare is $W_v(Q; \omega) = (a(\omega) + v - c)Q - \frac{K}{2}Q^2$. The risk-neutral first-best therefore replaces $\mathbb{E}[A]$ with $\mathbb{E}[A] + v$ in (37): $Q_{\text{FB}}^* = (\mathbb{E}[A] + v)/K$, establishing (i).

Step 2: The committing agent's uncorrected choice. The mean-CVaR agent who does not internalize v chooses $Q = A_{\text{eff}}/K$ with A_{eff} as in (47): the state-independent constant v appears in neither the market expectation nor the market tail mean she prices.

Step 3: Restoring credit. A per-unit credit σ_{tot} enters the committing agent's marginal return identically to a price increment (Step 5 of the proof of Theorem 1), so the credit that delivers Q_{FB}^* solves $A_{\text{eff}} + \sigma_{\text{tot}} = \mathbb{E}[A] + v$. Substituting (47),

$$\sigma_{\text{tot}} = v + \lambda m_\alpha (a_H - a_L) = v + \sigma^*(\alpha),$$

establishing (ii). For (iii), $\partial\sigma^*(\alpha)/\partial v = 0$ because v cancels from $\mathbb{E}[A] - A_{\text{eff}}$, and $\partial v/\partial(\lambda, \alpha, \pi_H, a_H - a_L) = 0$ because v is a primitive of the externality, not of the price distribution. No cross term appears because (47) is affine in v with unit coefficient. □

A.6. Proposition 2 and Corollary 4 (instrument form)

PROOF OF PROPOSITION 2. *Part (i): flat credit.* Immediate from Step 5 of the proof of Theorem 1: the credit $\sigma^*(\alpha)$ restores Q_{RN}^* , and it is paid on every unit in every state, so the expected per-unit outlay is $T_{\text{flat}} = \sigma^*(\alpha)$.

Part (ii): one-sided floor. Fix a strike $\kappa \in [a_L, a_H)$. The floor pays $\kappa - a(\omega)$ per unit exactly when $a(\omega) < \kappa$, i.e. in state L only, so the post-floor economy is the two-state economy with intercepts (a_H, κ) , unchanged probabilities, and unchanged state ordering ($\kappa < a_H$). The regime classification in Step 2 of the proof of Theorem 1 depends only on (α, π_H) , so (47) applies with the floored primitives:

$$A_{\text{eff}}^{\text{fl}}(\kappa) = \mathbb{E}[A] + \pi_L(\kappa - a_L) - \lambda m_\alpha (a_H - \kappa),$$

using $\mathbb{E}^{\text{fl}}[A] = \pi_H(a_H - c) + \pi_L(\kappa - c) = \mathbb{E}[A] + \pi_L(\kappa - a_L)$. The restoring condition $A_{\text{eff}}^{\text{fl}}(\kappa^*) = \mathbb{E}[A]$ reads

$$(48) \quad \pi_L(\kappa^* - a_L) = \lambda m_\alpha (a_H - \kappa^*),$$

whose unique solution is the convex combination

$$\kappa^* = \frac{\pi_L a_L + \lambda m_\alpha a_H}{\pi_L + \lambda m_\alpha},$$

which lies in (a_L, a_H) whenever $\lambda m_\alpha > 0$ (and equals a_L , a vacuous floor, at $\lambda = 0$, consistent with $\sigma^* = 0$). The expected per-unit outlay is the state- L payment weighted by its probability:

$$T_{\text{floor}} = \pi_L(\kappa^* - a_L) = \frac{\pi_L \lambda m_\alpha (a_H - a_L)}{\pi_L + \lambda m_\alpha} = \sigma^*(\alpha) \cdot \frac{\pi_L}{\pi_L + \lambda m_\alpha},$$

where the middle equality substitutes $\kappa^* - a_L = \lambda m_\alpha (a_H - a_L) / (\pi_L + \lambda m_\alpha)$ from (48).

Part (iii): two-sided CfD. Under the swap at strike $\bar{a} = \mathbb{E}[a(\omega)]$, the per-state, per-unit margin becomes $a(\omega) + (\bar{a} - a(\omega)) - c = \bar{a} - c$, deterministic. For a degenerate distribution, $\text{CVaR}_\alpha[W] = \mathbb{E}[W] = W$ at every $\alpha \in (0, 1)$, so the mean-CVaR objective collapses to $(\bar{a} - c)Q - \frac{K}{2}Q^2$ for every $\lambda \in [0, 1]$, with maximizer $Q = (\bar{a} - c)/K = \mathbb{E}[A]/K = Q_{\text{RN}}^*$. The expected per-unit outlay is $T_{\text{CfD}} = \mathbb{E}[\bar{a} - a(\omega)] = 0$ by construction.

Ranking. For $\lambda \in (0, 1]$: $m_\alpha > 0$ under Assumptions 1 and 6, so $T_{\text{floor}} > 0 = T_{\text{CfD}}$ and $T_{\text{floor}}/T_{\text{flat}} = \pi_L/(\pi_L + \lambda m_\alpha) \in (0, 1)$, giving $0 < T_{\text{floor}} < T_{\text{flat}}$. $\square$

PROOF OF COROLLARY 4. Value net public transfers state-by-state at a *state-independent* shadow price of public funds $\mu \geq 1$ (a constant marginal cost of public funds is maintained throughout; a state-dependent $\mu(\omega)$ would add a covariance term to the two-sided CfD's deadweight cost and is outside the corollary's scope): a transfer of $t(\omega)$ per unit to the chain in state ω costs society $(\mu - 1)t(\omega)$ per unit beyond the transfer itself (a negative $t(\omega)$ — the high-state leg of the two-sided CfD — is a receipt and carries the symmetric shadow saving). All three instruments in Proposition 2 induce the same output Q_{RN}^* and the same market surplus, so welfare under instrument i differs from welfare under instrument j only through the expected deadweight term:

$$\Delta W_{ij} = (\mu - 1)(T_j - T_i) Q_{\text{RN}}^*,$$

by linearity of the expectation in $t(\omega)$. At $\mu = 1$ the term vanishes and the instruments are welfare-equivalent; at

$\mu > 1$ the ranking of Proposition 2 becomes strict in welfare. $\square$

B. Aggregation Invariance

PROPOSITION 3 (Aggregation invariance of the harvest-cost term). *Let the hex-level Stage-1 objective use harvest-cost coefficient $c_h = \kappa \cdot \text{HC} / (\text{AvailInv}_h + \varepsilon)$, where $\kappa, \text{HC} > 0$, $\varepsilon \geq 0$, and AvailInv_h is hex h 's available Year-0 residue inventory. Define county aggregates $\text{AvailInv}_F = \sum_h \theta_{h,F} \cdot \text{AvailInv}_h$ and $K_F = \sum_h \theta_{h,F}$, where $\theta_{h,F}$ is the area share of hex h in county F with $\sum_F \theta_{h,F} = 1$, and set $c_F = \kappa \cdot \text{HC} \cdot K_F / (\text{AvailInv}_F + \varepsilon)$. Assume (A1) harvest allocations are proportional to availability with a common factor, $\bar{x}_h = \gamma \text{AvailInv}_h$ and $\bar{x}_F = \gamma \text{AvailInv}_F$ for some $\gamma > 0$ (the unconstrained Stage-1 first-order condition); and (A2) $\varepsilon = 0$, or $\text{AvailInv} \gg \varepsilon$ at every node. Then the harvest-cost totals coincide: $\sum_F c_F \cdot \bar{x}_F = \sum_h c_h \cdot \bar{x}_h$ (exactly under $\varepsilon = 0$; up to $O(\varepsilon)$ otherwise).*

PROOF. Under (A1) and $\varepsilon = 0$, $c_h \bar{x}_h = \kappa \text{HC} \gamma$ for every hex, so $\sum_h c_h \bar{x}_h = \kappa \text{HC} \gamma |H|$. At the county level, $c_F \bar{x}_F = \kappa \text{HC} \gamma K_F$, and $\sum_F K_F = \sum_F \sum_h \theta_{h,F} = \sum_h \sum_F \theta_{h,F} = |H|$, using $\sum_F \theta_{h,F} = 1$. Hence $\sum_F c_F \bar{x}_F = \kappa \text{HC} \gamma |H| = \sum_h c_h \bar{x}_h$. For $\varepsilon > 0$ the two sides differ by a Jensen-type term in the within-county dispersion of AvailInv , which is $O(\varepsilon)$ under (A2) and vanishes as $\varepsilon \rightarrow 0$. $\square$

Scope. The proposition covers the harvest-cost term only. Hypothesis (A1) is the unconstrained Stage-1 first-order condition; it fails where depot minimum-throughput, capacity, or ash-share constraints bind, as they do in the Stage-2 MILP. Transport aggregation uses the distance-averaging rules below, a linearization rather than an identity; revenue, OPEX, CAPEX, and the consumer-surplus integral aggregate by summation. The operative bound for the full accounting stack is therefore the empirical residual of $\pm 0.15\%$ reported in Table 7, not the exact identity.

Construction. The Stage-2 operational solve runs at the FIPS-5 county level rather than the underlying 50 mi^2 hex resolution. Because 8,864 of 14,040 hexes straddle multiple counties, hex-level quantities are split across counties proportional to area shares $\alpha_{h,F} \equiv \text{area}(h \cap F) / \text{area}(h)$, with $\sum_F \alpha_{h,F} = 1$ for every hex h . The aggregation rules for the five Stage-1 fields are:

$$\begin{aligned}
 \text{inv}_F[w] &= \sum_h \alpha_{h,F} \cdot \text{inv}_h[w] && \text{(area-attribute)} \\
 \text{growth}_F[w] &= \sum_h \alpha_{h,F} \cdot \text{growth}_h[w] && \text{(area-attribute)} \\
 \text{ratio}_F &= \frac{\sum_h \alpha_{h,F} \cdot \text{ratio}_h \cdot \text{inv}_h^{\text{HW}}}{\sum_h \alpha_{h,F} \cdot \text{inv}_h^{\text{HW}}} && \text{(HW-inventory-weighted)} \\
 \text{dem}_F[w] &= \sum_{m: \text{maj}(m)=F} \text{dem}_m[w] && \text{(point-facility majority)} \\
 d_{F,F'}^{\text{HM}} &= \frac{\sum_{(h,m): \text{maj}(m)=F'} d_{h,m} \cdot \alpha_{h,F}}{\sum_{(h,m): \text{maj}(m)=F'} \alpha_{h,F}} && \text{(hex-share + mill-majority)}
 \end{aligned}$$

The key step is the K-factor cost normalization: the hex-level Stage-1 LP objective coefficient is $c_{h,w} = \text{HC}_w \cdot 6.25 \cdot 3200 / (\text{AvailInv}_{h,w}^{Y_0} + \varepsilon)$; naively aggregating the denominator across hexes shrinks the objective by

Table 7: Hex-level vs. county-level totals at the deterministic baseline

Metric	Hex-level	County-level	Relative diff.
Total NPV cost (10-yr)	\$98.21 B	\$98.36 B	+0.15%
Total AResidue (pre-SRU, 10-yr cum.) ^a	1.205 B dt	1.204 B dt	−0.07%
Total Stumpage TSC	\$29.95 B	\$29.95 B	0.00%

Notes: Aggregation invariance holds within 0.2% across all three accounting metrics.

^a The AResidue row is a hex-vs-county aggregation-invariance check computed *before* the 0.70 sustainable-removal scaling, as a 10-year cumulative of a growing series; it is not a supply statement. On a like-for-like post-SRU basis the comparison is 0.844 vs. 0.843 B dt (−0.07%; the common factor cancels from the relative difference). The operational Stage-2 model consumes Year-0 availability only: 72.5 M dt (Section 4).

$\sim K$ = avg hexes/county. We restore invariance by setting

$$c_{F,w} = HC_w \cdot \frac{6.25 \cdot 3200 \cdot K_F}{\text{AvailInv}_{F,w}^{Y_0} + \epsilon}, \quad K_F \equiv \sum_{h \in P} \alpha_{h,F}.$$

Under the LP-optimal first-order condition $x_F[w] \propto \text{AvailInv}_{F,w}$, this gives $\sum_F c_{F,w} x_F[w] = \sum_h c_{h,w} x_h[w]$, so county-level and hex-level total costs coincide.

Feedstock envelope. Two envelope parameters referenced in Section 4 are fixed as follows. Transport catchments are capped at $H2D_{\max} = D2R_{\max} = 100$ mi for the source-to-depot and depot-to-refinery legs respectively. The source-to-depot leg is widened from the 50-mi convention of the antecedent siting literature because the sustainable-removal cut below shrinks per-county recoverable residue, and the 200 kdt/yr depot minimum-throughput constraint requires a wider draw to remain feasible at Grand Challenge scale; the depot-to-refinery leg is widened from 75 mi to 100 mi for symmetry with the first leg. Recoverable residue is scaled by a sustainable-removal fraction of 0.70, the U.S. DOE biomass scoping convention (U.S. Department of Energy 2016), applied on top of the willingness-to-harvest ($W2HF = 0.28$) and commercial-private share ($\text{ComR} = 0.338$) filters of Section 4.

Numerical sanity check. At the deterministic baseline ($p_{\text{SAF}} = \$5.97/\text{gal}$, 10-year horizon), the hex vs. county discrepancies are:

Per-state breakdowns can shift by up to $\pm 10\%$ (the LP re-allocates harvest within state to exploit county-level cost variation), but the total and the state-ranked Top-3 are preserved.

Table 8 records the dimensions of the aggregated instance that the rest of the paper solves.

C. KDE Discretization

We fit a Gaussian kernel density estimator (KDE) on the constructed drop-in biofuel price series described in Section 4: 3,046 daily-interpolated observations from weekly sources, August 2013 through December 2021.

Construction and provenance. The series is built in three steps, inherited from the producer-revenue calibration of Li et al. (2026). (i) Weekly U.S. retail gasoline and diesel prices (U.S. Energy Information Administration

Table 8: Spatial and temporal scope of the SE-U.S. residue-to-SAF supply-chain MILP

Quantity	Value
<i>Spatial frame</i>	
Hexagons (50 mi ² , SE-US frame)	14,040
Counties (FIPS-5, after hex → county agg.)	1,374
Mills $ \mathcal{M} $ (Stage-1 wood demand)	1,663
Active residue-source counties $ \mathcal{H} $ (Year-0, hardwood)	220
Depot candidate counties $ \mathcal{D} $ (528 industrial hexes)	483
Refinery candidate counties $ \mathcal{R} $	483
<i>Eligible edge sets (operational cutoffs, county level)</i>	
Stage-1 hex → mill pairs (≤ 100 mi)	244,634
Stage-2 residue → depot pairs (≤ 100 mi)	6,600
Stage-2 depot → refinery pairs (≤ 100 mi)	11,314
<i>Time and scenarios</i>	
Planning horizon (Year ₀ . . . Year ₉)	10 yrs
SAF-price scenarios $ \mathcal{S} $ (KDE bins)	20
Ash-tier categories ($A_d \times A_r$)	10×10

Notes: Hex resolution is 50 mi² (U.S. Forest Service, USFS, Wood Resources frame); the operational Stage-2 solve aggregates to the FIPS-5 county level (five-digit Federal Information Processing Standards codes). Both Stage-2 catchment legs are capped at 100 mi (widened from the 50/75-mi hex-model convention so that depot minimum-throughput constraints remain feasible at county resolution; Section B). Pair counts are county-level edge counts after the cutoff is imposed. Variable and constraint counts, the MIP-gap target, and realized gaps are reported separately in Table 11.

2024) are slate-weighted and adjusted to a common real basis by a compounded real-interest parameter, giving the fuel-market component (sample mean \$3.10/gal). (ii) Weekly D-code RIN prices are scaled by the 1.7 cellulosic equivalence factor and the same real adjustment, giving the RFS compliance-value component (sample mean \$2.96/gal); renewable jet equivalence values are in practice assigned facility-by-facility at 1.6–1.7 with most at 1.6 (Section E), and re-scaling to 1.6 would move the mean to $\approx$ \$5.88/gal, within the robustness of every downstream object. (iii) The revenue price is their sum, mean \$5.98/gal with standard deviation \$1.27/gal. The raw workbook, construction script, and merged series are part of the replication package with source identifiers and download dates. Two properties of the construction are load-bearing and are stated here rather than left for the reader to discover. First, the price is *policy-inclusive*: roughly half of its level is RFS compliance value, so the non-capitalization hypothesis of Proposition 1 fails with respect to RFS support, and Section 5.5 re-states the implied-SCC inversion accordingly (the baseline reading is *incremental to the existing RIN stack*). Second, the dispersion that the risk objects price is genuine producer revenue risk: a producer paid partly in credits bears RIN volatility one-for-one, and the two-state spread $a_H - a_L = \$2.19/\text{gal}$ reflects joint fuel-price and credit-price variation over the sample. The weekly source data are serially persistent and the daily interpolation adds no information; refitting the KDE on the 436 weekly source observations moves the collapse primitives little ($\pi_H = 0.44$, spread \$2.20, $\sigma^* = \$0.48$ against the published \$0.52), and halving or doubling the Silverman bandwidth moves σ^* only within \$0.48–\$0.53. Persistence matters more than frequency. The scenario structure

of the two-stage program treats the realized price as a persistent decade-scale state, the reading under which the spot dispersion (spread \$2.19) is the relevant exposure. Under the opposite extreme — rapid mixing, so the committed chain earns the multi-year *average* price — a moving-block bootstrap of H -year averages (26-week blocks, 2,000 draws) gives spreads of \$1.34, \$0.58, and \$0.41 at $H = 1, 5, 10$ and σ^* of \$0.35, \$0.14, and \$0.11. The risk component therefore carries the persistence assumption openly: \$0.52 under permanent price states, falling toward \$0.11 under full decade-scale mean reversion. An 8.4-year sample cannot pin the long-run persistence of the fuel-plus-RIN price level, and we flag this as the primary calibration uncertainty on the risk side (Section 6.4).

Identification: mapping the constructed price to SAF revenue. SAF is not yet a deep liquid market: in 2026, the modal SAF-buyer is an airline operating under bespoke ESG offtake contracts (United-NEXT, Delta–Air-Products, Microsoft–Twelve) at premia of approximately \$2–\$4/gal above wholesale jet-A. The constructed series proxies the revenue of a drop-in fuel producer paid the fuel-market price plus compliance-credit value; for SAF the corresponding stack is jet-A plus the credit-and-premium package, of comparable magnitude to the constructed mean. The slack-import term $p^{\text{jet}} \cdot s(\omega)$ in (3) remains a fossil-jet benchmark on a policy-free basis (Section 2.2), so the make-or-buy comparison prices domestic production inclusive of its policy revenue against fossil import at its market price; the welfare interpretation of the RIN component — transfer versus resource value — is taken up where the decomposition uses the price level (Section 5.5). Two first-order robustness notes: σ^* depends on the state spread $a_H - a_L$, which is robust to additive shifts of the level; the EVPI level scales linearly with $\text{Var}(p_{\text{SAF}})$ and would shift with the (unobserved) variance of a mature SAF market relative to the constructed series.

Evaluation vintages, certification, and convergence. The per-scenario statistics of Section 5.2 come from a certified evaluation computed under a single data state. Two vintages preceded it, and each correction moved the estimate the same way. (i) An early vintage — retained in the replication package — carried low-price wait-and-see gaps of 27–79% and required a dominance-certificate correction (raw $\mathbb{E}[L] = \$97.7$ M/yr versus conservative \$169 M/yr). (ii) A refreshed vintage re-solved the six lowest-price bins under a 7,200 s limit with small-topology warm starts (realized gaps 1.3–5.0%) and gave $\mathbb{E}[L] = \$185$ M/yr; but its own candidate evaluations exposed the incumbency trap one level up: the 197/27 plan’s fixed-siting values exceeded the recorded wait-and-see incumbents in bins 2 and 4–9, by \$48 M to \$432 M — negative raw regrets that are certificates of wait-and-see suboptimality, not noise to be floored away. (iii) The certified vintage re-solves those seven bins warm-started from the certifying topology; the recorded wait-and-see values now dominate every fixed-siting evaluation in the candidate pool, realized gaps are 1.3–3.9%, and $\mathbb{E}[L] = \$299$ M/yr; every here-and-now value quoted in the text is a fixed-siting solve with recorded gap below 10^{-12} . Each step raises the floor of a lower-bound band whose gap-implied ceiling is \$827 M/yr (Section 5.2). Known artifacts of the evaluation files — including one superseded candidate whose rows duplicate the central plan and are excluded from every statistic — are documented in the replication package.

Calibration choice for the regret evaluation. An earlier evaluation paired the elastic-side anchor $\varepsilon = -1.0$ with a \$5.68/gal voluntary-premium anchor. At that calibration the low-price wait-and-see MILPs do not improve on the warm start within the time budget — the bin-1 solve terminates at a 79% residual gap — and the recorded regret lower bound degenerates to \$4.5 M/yr, a convergence artifact rather than an economic finding. The $\varepsilon = -0.5$

Table 9: The candidate pool of the risk evaluation: provenance and per-plan statistics at the refined 81-breakpoint grid

Plan	Bin	Vint.	N_D/N_R	Prov. gap	$\mathbb{E}[L]$	CVaR ₉₅ [L] (\$M/yr)	$\mathbb{E}[W]$	CVaR ₉₅ [W] (\$B/yr)
committed ^a	9	(i)	248/34	18.12% ^b	300	931	17.004	11.426
tight01	1	(ii)	192/26	2.04%	626	1,804	16.678	12.351
tight02	2	(ii)	195/26	3.86%	949	2,337	16.355	12.134
tight03	3	(ii)	197/27	1.34%	400	1,837	16.904	12.353
tight04	4	(ii)	204/28	2.14%	369	2,062	16.935	12.197
tight05	5	(ii)	211/30	3.91%	508	1,334	16.797	11.959

Notes: Statistics from LP-exact fixed-siting recourse evaluations of each plan across all 20 KDE bins at an 81-breakpoint SOS2 grid, against the pool-maximum wait-and-see column (Section 3.5); the 21-breakpoint evaluation’s cross-plan margins are chord-error artifacts (Section 5.8). Vintage labels follow the evaluation-history paragraph above. The committed plan minimizes $\mathbb{E}[L]$ and CVaR₉₅[L] and maximizes $\mathbb{E}[W]$; tight03 maximizes CVaR₉₅[W]; plans 195/26, 204/28, and 211/30 are no longer scenario-optimal after certification and are retained as candidates only. The vintage-(ii) bin-6 solve is bit-identical to the committed plan and is not listed separately.

^a Provenance: deterministic bin-9 sweep; the identical siting is reproduced by the bin-6 endogenous-demand solve at 4.81%.

^b On the fixed-demand PVC objective of the provenance sweep.

evaluation reported in Section 5.2 is the one whose low-price solves move far enough off the warm start to be informative, including the certified low-price re-solves of vintage (iii) above.

The KDE bandwidth follows Silverman’s rule of thumb (Silverman 1986),

$$h^* = 1.06 \cdot \hat{\sigma} \cdot n^{-1/5},$$

robust to mild non-Gaussianity in this constructed price series. Sheather–Jones (Sheather and Jones 1991) and Wand–Jones (Wand and Jones 1995) plug-in alternatives shift the bandwidth by $\pm 7\%$; the 20-bin discretization absorbs this variation (σ_{KDE}^* moves by $< 3\%$ across the three bandwidth rules).

We discretize the continuous KDE into 20 equal-width bins over the empirical support [\$3.81, \$9.12]/gal, assigning each bin its KDE-weighted probability p^ω and conditional mean $p_{\text{SAF}}(\omega) = \mathbb{E}[p_{\text{SAF}} \mid p_{\text{SAF}} \in \text{bin}_\omega]$ (Table 2). The raw monthly sample has mean \$5.97/gal and standard deviation \$1.27/gal; the 20-bin discretization has probability-weighted mean $\mathbb{E}[p_{\text{SAF}}] = \$6.00/\text{gal}$ and standard deviation \$1.28/gal, and all discretization-derived quantities in the paper use the binned values. The bin count of 20 matches the typical precision-vs.-tractability trade-off in scenario-based SP (Birge and Louveaux 2011) and exactly resolves the empirically relevant tail at $\alpha = 0.95$ (cf. Table 2: $p^{19} + p^{20} = 0.012$, which sits cleanly within the $1 - \alpha = 0.05$ CVaR confidence window).

Bin-count sensitivity. We test the robustness of the 20-bin choice by re-aggregating the KDE at $M \in \{10, 20, 40\}$ bins and recomputing the 2-state collapse of Section 2.4. At $M = 10$ (pair-merging consecutive bins), the KDE-collapsed $\sigma_{\text{KDE}}^* = \$0.518/\text{gal}$ and EVPI at $K = 1$ is \$0.600/gal², matching the $M = 20$ baseline values exactly because pair-merging preserves the median-split conditional means. At $M = 40$ (splitting each bin at its midpoint under uniform-within-bin densities), σ_{KDE}^* shifts to \$0.490/gal and EVPI to \$0.610/gal² — a 5.4% drop in σ^* and a 1.7% rise in EVPI relative to baseline. The 20-bin choice is therefore a stable representative of

Table 10: Per-bin certified wait-and-see values, realized gaps, and gap-implied regret allowance (endogenous-demand evaluation)

Bin	p^ω	W^{WS} (\$B)	gap	L_s (\$M)	allowance $p^\omega g_\omega W^{\text{WS}}$ (\$M)
1	0.0328	12147914935.756	2.0%	947	8.1
2	0.0593	12723909183.871	2.9%	872	21.5
3	0.0750	13308753337.244	1.3%	781	13.4
4	0.0749	13925039652.988	2.8%	671	29.2
5	0.0675	14579488394.909	2.9%	554	28.4
6	0.0671	15269195972.772	3.0%	432	31.0
7	0.0743	15877584179.751	3.2%	325	37.5
8	0.0772	16578743018.242	3.5%	180	44.7
9	0.0635	17118613056.083	3.9%	48	42.0
10	0.0444	17951363404.618	3.7%	0	29.5
11	0.0395	18755595944.940	3.3%	0	24.8
12	0.0481	19554799375.917	3.1%	0	29.5
13	0.0581	20253857812.329	3.1%	0	36.2
14	0.0607	21101838682.476	3.1%	0	39.1
15	0.0565	21868292880.931	3.1%	0	37.7
16	0.0454	22667461822.222	3.1%	0	31.8
17	0.0289	23429918450.411	3.1%	0	21.3
18	0.0148	24293572821.621	3.2%	0	11.7
19	0.0079	25023687891.312	3.5%	0	6.9
20	0.0042	25759765497.633	3.7%	0	4.0
Expectation					528

Notes: g_ω is the realized relative gap of the recorded wait-and-see value. The allowance column sums to the \$528M gap-implied expectation quoted in Section 5.2; because the certified wait-and-see column coincides with the candidate-pool maximum in every bin, the allowance in bins 10–20 is capacity for unrecorded regret outside the pool, not evidence of it.

the KDE for both headline quantities.

D. Computational Details

Source runs for the committed plan. The committed here-and-now reference of Section 5 is the bin-9 central-scenario solve (248 depots / 34 refineries, 3.32 B gal/yr, \$1.43 B/yr chain surplus); a companion re-solve at the raw mean price of \$5.97/gal places 228/31 (3.07 B gal/yr, \$1.20 B/yr). The \$0.08/gal difference between the two calibrations straddles the topology margin at the lower edge of the expansion band in Figure 8, which is why the two runs differ in siting. Both terminate at the wall-clock limit with realized relative gaps of 18.12% (bin 9) and 15.20% (\$5.97), so the reported topologies are feasible incumbents rather than certified optima; both runs are retained in the replication package.

The Stage-2 problem is solved separately for each of the 20 KDE scenarios in the candidate-evaluation spirit of Birge and Louveaux (2011) (Ch. 4). The Mak, Morton, and Wood (1999) and Bayraksan and Morton (2011) replication-based schemes would deliver a *statistical* upper bound on $\mathbb{E}[W^*]$ via Monte Carlo replications of the scenario set; we do not run replications, and the deterministic 20-bin discretization does not by itself deliver a certified bound either, since the per-scenario values entering the expectation are incumbents rather than best bounds. What the reported \$255 M estimate is — and is not — is set out at the end of this appendix.

Table 11: Computational profile of the Stage-1 LP and Stage-2 MILP at the baseline configuration. The Stage-2 column refers to the cumulative 20-scenario wait-and-see sweep behind the chain-surplus information-value accounting of Section 5.3; per-scenario solves are run at MIP-gap target 0.05; the bin-9 baseline solve realizes an 18.12% gap and the sweep median is 8.07% (the endogenous-demand regret statistics of Section 5.2 come from the separate evaluation of Section 3.5).

Metric	Stage-1 LP	Stage-2 MILP
Continuous variables	489,268	~700,000
Binary variables	0	~10,000
Constraint rows	23,820	~50,000
Non-zeros in constraint matrix	689,597	~3.5 M
MIP gap (5% target)	exact	18.1% realized (baseline)

The elasticity $\times$ operating-cost grid. Table 12 reports the nine-cell grid of Section 4.4 together with the build-vs-slack crossing at each operating cost. The crossing is a fixed-demand object, so it does not vary with the elasticity and is constant down each OPEX column; the three distinct values are obtained by re-solving the crossing-region bins at \$1.96, \$2.30, and \$2.70/gal rather than by shifting the baseline crossing one-for-one with operating cost. That shortcut turns out to understate the move: the measured crossings are \$5.91–\$6.06 and \$6.31–\$6.44 against the \$5.8 and \$6.2 a one-for-one shift would give, so the crossing rises somewhat faster than the operating-cost assumption that drives it.

Per-scenario realized gaps and asymmetric-gap argument. The 20-bin wait-and-see sweep is solved at a 5% per-scenario relative-gap target; realized gaps cluster around the target but are not uniformly at 5% (Table 13). The median realized gap is 8.07% with inter-quartile range [6.01%, 11.91%], and the maximum realized gap (excluding bin 7, where $PVC^\omega \approx 0$ mechanically inflates the relative metric) is 21.18%.

The near-zero PVC^ω at bin 7 reflects build-vs-slack indifference at the marginal SAF wholesale price: at $p_{SAF} = \$5.53/\text{gal}$, the per-gallon welfare contribution of domestic SAF (revenue minus chain cost) is approximately the per-gallon opportunity cost of slack imports ($p^{\text{jet}} = \$3.50/\text{gal}$), so the planner is approximately indifferent between building the marginal facility and purchasing the slack gallon. Bin 7 sits at the structural-equilibrium pivot of the deterministic single-scenario problem, and the relative-gap inflation is a numerical artifact of this indifference, not a methodological failure; the absolute gap \$0.14 B remains within the per-scenario time budget.

Let UB_ω and LB_ω denote, respectively, the solver-reported best-bound and incumbent objective values for wait-and-see scenario ω , and let $a_\omega = UB_\omega - LB_\omega \geq 0$ be the realized absolute gap (the relative gap $g_\omega = a_\omega/|LB_\omega|$ is reported for reference but is sign-unsafe when $LB_\omega < 0$, as at bin 7). By construction $UB_\omega = LB_\omega + a_\omega$ for every ω . Forming $\mathbb{E}[W^*]_{UB} \equiv \sum_\omega p^\omega UB_\omega$ yields an *upper* bound on the true wait-and-see expectation; the corresponding lower bound $\mathbb{E}[W^*]_{LB} = \sum_\omega p^\omega LB_\omega$ would still be valid but does not control the sum of LB_ω from above. What the reported number actually pairs, however, is $\mathbb{E}[W^*]_{LB} = \sum_\omega p^\omega LB_\omega$ — the *incumbent-weighted* wait-and-see expectation, a lower bound on the true $\mathbb{E}[W^*]$ — with $\max_c \mathbb{E}[W | y_c]$, the best-candidate evaluation, itself a lower bound on the true recourse optimum $W^{\text{RP}*}$ (the extensive-form recourse problem was not solved). A certified one-sided bound of the form $\mathbb{E}[W^*]_{UB} - W^{\text{RP}}_{LB}$ would require the best-bound-weighted expectation $\sum_\omega p^\omega UB_\omega$, which the gap-target budget did not deliver at the low-price bins

Table 12: Nine-cell elasticity $\times$ operating-cost robustness grid

ε	OPEX (\$/gal)	Depots N_D	Refineries N_R	Realized gap (%)	Crossing (\$/gal)
−0.3	1.96	248	34	3.3	5.49–5.54
−0.3	2.30	248	34	3.7	5.91–6.07
−0.3	2.70	248	34	4.1	6.31–6.44
−0.5	1.96	248	34	4.9	5.49–5.54
−0.5	2.30	248	34	5.1	5.91–6.07
−0.5	2.70	248	34	6.4	6.31–6.44
−1.0	1.96	248	34	7.6	5.49–5.54
−1.0	2.30	248	34	11.0	5.91–6.07
−1.0	2.70	185	27	10.6	6.31–6.44

Each cell is a Path B endogenous-demand re-solve at a 5% gap target, warm-started from the nearest cell already solved; the \$2.30 and \$2.70 columns price an H₂-inclusive configuration (Section 4.4). Bold marks the one cell that departs from the 248/34 first-stage topology; at a 10.6% realized gap it is an unconverged incumbent rather than a certified topology shift. The build-vs-slack crossing is a fixed-demand object and carries no elasticity, so it is constant down each OPEX column. It is reported as an interval, interpolated once from the incumbents and once from the bounds of re-solves of the crossing-region bins at that operating cost: several of those solves stop well short of the gap target, but every bin used to bracket a crossing has its incumbent and its bound on the same side of zero, so its position relative to the crossing is certain and only the interpolated point inherits the gap. Relative gaps are uninformative for bins whose surplus is near zero – the \$2.30 bin at \$6.05/gal reports 817% on an absolute spread of \$0.44 B – and that bin is also where the topology expands from 226/31 to 248/34, so the build threshold and the expansion point nearly coincide.

(median realized gap 8.07%). Because the difference of two lower bounds has no determinate bias sign, we report \$255 M as a *candidate-based estimate* of the chain-surplus EVPI — correcting the “one-sided upper bound” reading of earlier drafts, which described a construction that was not the one computed. The same argument shuts down VSS reporting, since W^{EEV} requires solving a separate MILP that carries the same gap asymmetry without a candidate-evaluation upper bound to anchor it (Birge and Louveaux 2011).

E. Section 45Z: Institutional Detail

This appendix records the institutional machinery behind the §45Z clean fuel production credit that our decomposition takes as its policy benchmark, and the respects in which certified credit values could differ from the GREET/CORSIA-derived carbon intensities used in the model.

Carbon-intensity certification. The credit equals an applicable amount times an emissions factor, $(50 - \text{ER})/50$, where ER is the fuel’s lifecycle emissions rate in kgCO₂e per mmBtu (United States Code 2025a). For fuel produced from 2025 onward, Treasury guidance (Notices 2025-10 and 2025-11, issued January 10, 2025) directs producers to the 45ZCF-GREET model, a purpose-built derivative of Argonne’s GREET model published by the U.S. Department of Energy on January 15, 2025 and updated in May 2025 (Internal Revenue Service 2025a,b; U.S. Department of Energy 2025). The model covers gasification–Fischer–Tropsch pathways, but industry comment letters and trade analyses indicate that forestry and logging residues are not among its listed feedstocks, so a forest-residue FT producer must instead petition for a provisional emissions rate (PER): the

Table 13: Per-scenario realized MIP gaps on the 20-bin wait-and-see sweep (fixed-demand PVC accounting)

Statistic	Relative gap (%)	Absolute gap (\$B)	Realized PVC ^ω (\$B)
Min (excl. bin 7)	4.75	0.02	–
Median (excl. bin 7)	8.07	0.13	–
Max (excl. bin 7)	21.18	0.46	–
Bin 7 (separately)	455.80 [†]	0.14	–0.03

Notes: Summary statistics exclude bin 7 ($p_{\text{SAF}} = \$5.53/\text{gal}$), whose near-zero PVC^ω = –\$0.03 B mechanically inflates the relative gap (455.8%); its \$0.14 B absolute gap remains within the per-scenario time budget and is not anomalous in absolute terms ([†]).

producer first obtains an “emissions value” from DOE (requiring a front-end engineering and design study; DOE published its notice of intent for this process on July 24, 2025), then files a PER petition with the IRS; Treasury has indicated credits may be claimed retroactively to January 1, 2025 once a rate is established ([RSM US LLP 2025a](#); [Strategic Biofuels LLC 2025](#); [Advanced Biofuels USA 2025](#)). Our carbon-intensity range of 8.6–12.9 kgCO₂e/mmBtu is GREET/CORSIA-derived; the certified 45ZCF-GREET or PER value could differ, since taxpayers may not substitute their own background assumptions ([Holland & Knight LLP 2025](#)). At our range, the emissions factor is 0.74–0.83, so the credit per gallon is that fraction of the applicable amount below. Proposed regulations issued February 4, 2026 would further specify these rules ([U.S. Department of the Treasury and Internal Revenue Service 2026](#)).

Rates and the 2025 amendments. As enacted in the Inflation Reduction Act, §45Z paid a base rate of \$0.20/gal (\$1.00/gal if prevailing-wage and apprenticeship requirements are met), with a SAF premium rate of \$0.35/gal (\$1.75/gal) ([United States Code 2025a](#); [Congressional Research Service 2025a](#)). The 2025 budget reconciliation act (P.L. 119-21, enacted July 4, 2025) eliminated the SAF premium for fuel produced after December 31, 2025 — aligning SAF at the \$1.00/gal maximum — extended the credit through December 31, 2029 (from 2027), restricted eligible feedstocks to those produced or grown in the United States, Mexico, or Canada, excluded indirect land-use-change emissions from the emissions rate, and disallowed negative emissions rates except for animal-manure fuels ([RSM US LLP 2025b](#); [Clean Air Task Force 2025](#)). The dollar amounts are indexed for inflation for years after 2024 ([United States Code 2025a](#)). The §40B SAF credit that preceded 45Z (\$1.25–\$1.75/gal in 2023–24) expired December 31, 2024 ([Internal Revenue Service 2024](#)).

Transferability. Because SAF developers rarely have tax appetite in early operating years, most 45Z value is monetized by selling credits for cash to unrelated taxpayers under §6418 ([United States Code 2025b](#)). Market intelligence for 2025 reports transferable credits trading at 92–94 cents per dollar on average, with §45Z credits pricing *below* the market average — roughly 91 cents — reflecting their recent emergence, verification burden, and guidance risk ([Crux Climate 2025](#)). This discount is one concrete channel through which a statutory dollar of flat credit delivers less than a dollar of financeable revenue, whereas floor payments from a creditworthy counterparty enter lenders’ cash-flow models at face value.

Stacking with the RFS and state LCFS programs. FT jet fuel from forest residues qualifies under the Renewable Fuel Standard as cellulosic diesel (a category that includes jet fuel), generating D-code 7 RINs; equivalence

values for renewable jet fuel are assigned facility-by-facility under 40 C.F.R. §80.1415(c) and have historically been approved at 1.6–1.7 RINs per gallon, with most at 1.6 ([Code of Federal Regulations 2025a,b](#)). California, Oregon, and Washington treat alternative jet fuel as an *opt-in* fuel in their low-carbon fuel standards — SAF generates credits while conventional jet fuel generates no deficits — and these state credits stack with RINs and 45Z ([California Air Resources Board 2024](#); [Stillwater Associates 2025](#)). Within the tax code, stacking is restricted: no 45Z credit is allowed for a facility for which the §45V clean hydrogen or §45Q carbon sequestration credit is allowed in the same taxable year, determined year by year ([United States Code 2025a](#); [Baker Botts LLP 2025](#)). Capture differs across instruments: 45Z pays the registered *producer* (registration under §4101; sale to an unrelated party) ([Holland & Knight LLP 2025](#)), whereas the expired §40B credit attached to the qualified SAF *mixture* and was therefore captured at the blender level ([Internal Revenue Service 2024](#)); RINs are generated at production but separated and monetized upon blending, splitting value between producer and obligated party.

Incidence and risk-bearing. Statutory incidence of 45Z at the producer matters for our decomposition because the instrument’s risk-mitigation value depends on who holds the price exposure it is meant to offset. A flat per-gallon credit raises the mean of the producer’s revenue distribution but leaves its variance untouched; if offtake agreements pass price risk through to the producer (as in index-linked SAF offtakes), the risk component of the credit identified in Section 5 compensates the party that actually bears the tail risk, which is the efficiency argument for converting that component into a floor. If instead airlines absorb price risk through fixed-price offtakes, a producer-side flat credit over-delivers to the party bearing the least risk. The producer-level payment point of 45Z — unlike the blender-level 40B — therefore makes the floor conversion administratively natural: the same registered entity that certifies volumes and carbon intensity for the credit would be the counterparty of the floor contract.

F. Additional Figures

This appendix collects three supporting figures referenced in Section 5: the siting comparison behind the information-value accounting of Section 5.3 (Figure 6), the implied-SCC schedule behind the emissions-basis discussion of Section 5.5 (Figure 7), and the fixed-demand topological price sweep of Section 5.7 (Figure 8).

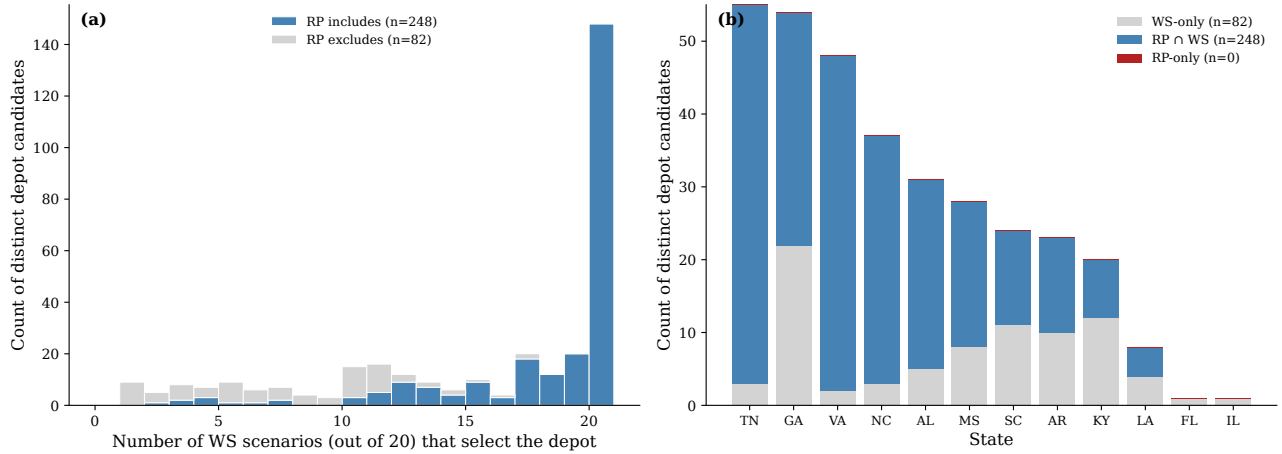


Figure 6: Siting comparison: recourse-problem first-stage solution (RP, 248 depots) vs. union of 20 wait-and-see per-scenario solutions (WS, 330 depots union). **Panel (a):** scenario-recurrence histogram — for each depot in the WS union, the number of WS scenarios (out of 20) in which the depot appears, separated by whether the RP solution includes the depot (steel blue) or excludes it (light gray). **Panel (b):** per-state stacked breakdown of WS-only / both / RP-only depot candidates across the SE-U.S. core and broader region.

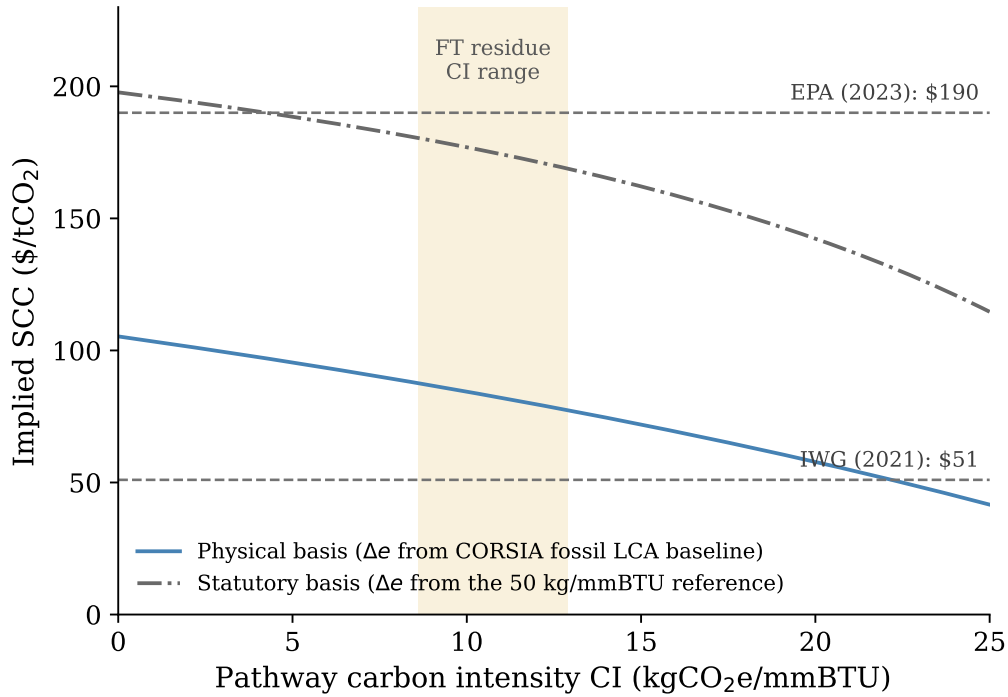


Figure 7: Implied SCC as a function of pathway carbon intensity. Both the statutory credit, $1.75 \times (50 - CI)/50$, and the emissions differential move with CI, so the implied SCC $(\text{credit}(CI) - \sigma^*)/\Delta e(CI)$ is a schedule. Solid: physical basis ($\Delta e = 11.70 - 0.1246 CI$ kgCO₂e/gal, CORSIA fossil life-cycle baseline). Dash-dotted: statutory basis ($\Delta e = 0.1246(50 - CI)$, the statute's 50 kgCO₂e/mmBTU reference). Shaded band: the FT-residue CI range 8.6–12.9 kgCO₂e/mmBTU consistent with the \$1.30–\$1.45/gal credit. The point band \$77–\$91 in (27) holds Δe fixed at 10.2 while varying the credit; the schedule here co-moves both, giving \$77–\$88 on the physical basis over the shaded range. Horizontal lines: the two federal SCC vintages.

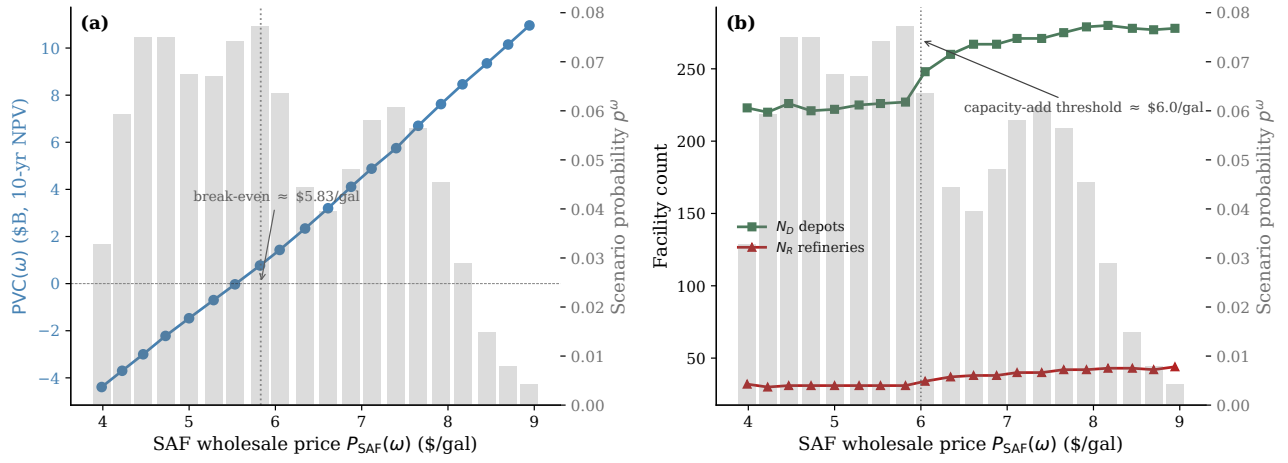


Figure 8: Topological response of the SE-U.S. residue-to-SAF supply chain to wholesale SAF price, across the 20 KDE-discretized scenarios (gray bars: scenario probabilities p^ω from Table 2). **Panel (a):** present-value chain surplus $\text{PVC}(\omega)$ vs. $p_{\text{SAF}}(\omega)$; the interpolated build-vs.-slack crossing is $\approx$ \$5.5/gal, and the dashed orange line marks the first positive-surplus bin (\$5.82). **Panel (b):** realized facility counts N_D, N_R vs. p_{SAF} ; the marginal-capacity expansion band spans roughly \$5.8–\$6.6/gal (dotted purple line at its lower edge), across which the chain adds depots and refineries quickly.